



INTERNATIONAL ROADMAP FOR DEVICES AND SYSTEMS™

2024 UPDATE

MORE MOORE

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MORE MOORE

1. INTRODUCTION

System scaling enabled by Moore's scaling is increasingly challenged by the scarcity of resources such as power and data bandwidth. This has become more challenging under the requirements of seamless interaction between big data and instant data (Figure MM-1). Instant data generation requires ultra-low-power devices with an "always-on" feature at the same time as high-performance devices that can generate the data instantly. Big data requires abundant computing, communication bandwidth, and memory resources to generate the service and information that clients need. Need for data next to compute accelerated as AI workloads increased.

The More Moore International Focus Team (IFT) of the International Roadmap of Devices and Systems (IRDS) provides physical, electrical, and reliability requirements for logic and memory technologies to sustain power, performance, area, cost (PPAC) scaling for edge and cloud applications. This is done over a time horizon of 15 years for mainstream/high-volume manufacturing (HVM).



Figure MM-1 Big data and instant data.

Following systems drivers are forecasted to impact the More Moore logic technologies:

Mobile

- Heterogenous integration
- Edge computing
- Extreme reality (VR/AR)
- AI enhanced edge computing and connectivity (mobile phone, 6G, cellular, IoT, Wi-Fi, wireless connectivity, smart cameras, and speakers) driving any data, any place, highest speed, and lowest power with content-rich data.

Data and HPC servers – cache integration, memory, IO

- AI accelerators in enterprise/cloud
- Codec ASICs - 24/7/365 continuous run of video and audio (codec), 5 years minimum time
- Networking – Always-on, 500W power envelope
- Advanced driver assistance system (ADAS) chips – Autonomous driving
- Memory and IO solutions for AI, graphics, HPC

Novel compute fabrics

- Neural processing unit

- Fine-pitch 3D stacking
- Reconfigurable compute fabrics
- Smart 2.5D interposers

1.1. CURRENT STATE OF TECHNOLOGY

A major portion of semiconductor device production is devoted to digital logic that needs to support a technology platform for two device types: 1) high-performance logic, and 2) low-power/high-density logic. Key considerations for this technology platform are speed, power, density, cost, capacity, and time-to-market. The More Moore roadmap provides an enablement view for continued scaling of MOSFETs to maintain historical trends of improved device performance at reduced power and cost and at high volume [1]. 2025 is the 100th anniversary of the concept of the Field Effect Transistor (FET), which was first captured as a patent filed by Julius Lilienfeld on October 22, 1925. While reaching the 100th anniversary of transistor, we have seen many critical enablers for the continuous enhancement of transistor on top of geometric scaling, such as strained silicon, high- κ metal gate, raised source drain, multi-gate transistor such as finFET, and now evolving to gate-all-around devices.

1.2. DRIVERS AND TECHNOLOGY TARGETS

The following applications drive the requirements of More Moore technologies that are addressed in the IRDS [2]:

- High-performance computing—more performance at constant power density (constrained by thermal)
- Mobile computing—more performance and functionality at constant energy (constrained by battery) and cost
- Autonomous sensing and computing (IoT)—targeting reduced leakage and variability

Technology drivers include the following focal items:

- Logic technologies
- Ground rule scaling
- Performance boosters
- Performance-power-area (PPA) scaling
- 3D integration
- Memory technologies
- DRAM technologies
- Flash technologies
- Emerging non-volatile-memory (NVM) technologies

More Moore targets bringing PPAC value for node scaling every 2–3 years [3]:

- (P)erformance: >10% more operating frequency at scaled supply voltage
- (P)ower: >20% less energy per switching at a given performance
- (A)rea: >30% less chip area footprint
- (C)ost: <30% more wafer cost – 15% less die cost for scaled die.

System scaling considers the co-integration of logic, memory and IO solutions bringing the following targets:

- TOPS (Tera Operations per Second): Throughput
- TOPS/W (TOPS per Watt): Energy efficiency
- TOPSxTOPS/W/Area is an indicator of Energy-Area-efficient performance (aka EDP: Energy Delay Product at Unit Area)
- 2.0-2.4x scaling of TOPSxTOPS/W/Area for node scaling per frame, per inference, per training, and/or per pocket

These scaling targets have driven the industry toward a number of major technological innovations, including material and process changes such as high- κ gate dielectrics and strain enhancement, and in the near future, new structures such as gate-all-around (GAA); alternate high-mobility channel materials, wafer backside processing (e.g. backside PDN) [4], and new

3D integration schemes allowing heterogeneous stacking/integration. These innovations will be introduced at a rapid pace, and hence understanding, modeling, and implementation into manufacturing in a timely manner is crucial for the industry.

It is important to note that both cost metric (15% less cost for the same function) and market cadence necessitating new products at high volume every year are becoming more important targets in the mobile and high-performance computing industry. As the applications strictly requiring all figure-of-merits (FoMs) are concurrently met, it is necessary to advance an effective list of process technologies for sustaining certain device architectures to their limits, such as pushing the gate-all-around architecture until 2031 while ensuring a swift transition to complementary-FET (CFET) devices and backside wafer processing, which will sustain more than another decade. This approach will also help in sustaining the cost at reduced risk while moving from one logic generation to another. This becomes more difficult whenever the cost of wafer processing becomes more expensive with the increased number of steps because of multiple patterning lithography steps. However, it is necessary to reduce the cost by more than 15% at every logic generation for the same number of transistors, which can only be enabled by pitch scaling enabled by new advancements in channel material, device architecture, contact engineering, device isolation, and most importantly by design-technology-co-optimization (DTCO). Increased process complexity must also be considered for the overall die yield. In order to compensate for the cost of complexity, acceleration in DTCO is needed to further scale the area to reach the die-cost scaling targets [3]. The DTCO scaling factors are now considered as one of the key elements in the More Moore technology roadmap.

2. SUMMARY AND KEY POINTS

The following are forecasted in the projected IRDS More Moore roadmap:

- Ground rule scaling is expected to slow down and saturate around 2031. High-NA Extreme-ultraviolet (EUV) technology will be the enabler of ground rule scaling while keeping the cost under control and providing process complexity reduction. Transition to 3D VLSI (dense logic-on-logic integration) after 2035.
- Ground-rule scaling needs to be accompanied by the design-technology-co-optimization (DTCO) constructs that accommodate the area reduction as well as tightening the critical design rules that limit the overall SoC area scaling.
- A main challenge in 3D integration is how to partition the system to produce better utilization of devices, interconnect, and sub-systems such as memory, analog, and input/output (I/O).
- Parasitics improvement will become a major knob for performance improvement, such as with the introduction of low- κ device spacer.
- 2D material channels have not reached manufacturing maturity and are expected to complement the platform devices with specialty devices.
- It becomes increasingly difficult to control interconnect resistance, electromigration (EM), and time-dependent-dielectric-breakdown (TDDB) limits. Interconnect resistance has now entered an exponential increase regime because of non-ideal scaling of barriers for Cu bringing less metallization volume and increased scattering at the surface and grain-boundary interfaces. Therefore, there is a need for new barrier materials, atomic layer deposition (ALD) based barrier deposition, and/or high aspect-ratio non-Cu metallization solutions, with the latter using direct metal etch. In addition to the resistance scalability, TDDB is putting a limit on the minimum space between the adjacent lines for a given low- κ dielectric, forcing a slow-down in the permittivity (κ -value) scaling and forces the introduction of air-gap in tight-pitch metals to cope with the capacitance at high aspect-ratio local interconnect.
- Performance is forecasted to improve on average for wireloaded datapaths, most of improvements taking place as transition from GAA to CFET devices.
- System-on-chip (SoC) level area across six nodes spanning from 2024 to 2039 is forecasted to improve, but less than 25%, node-to-node on average.
- Power density poses a significant challenge for scaling, particularly because of 3D integration after 2035. Therefore, it is necessary to factor in thermal considerations in devices and architectures.
- Energy per switching reduction significantly slowed down because of a slow-down in capacitance and supply voltage reduction.
- DRAM needs to maintain sufficient storage capacitance and adequate cell transistor performance is required to keep the retention time characteristic in the future. If efficiency of cost scaling becomes poor in comparison with introducing the new technology, DRAM scaling will be stopped and pillar structure adopting 4F2 cell structure and 3D DRAM cell stacking structure will be adopted. Alternatively, a new DRAM concept could be adopted.

- 2D Flash memory density cannot be increased indefinitely by continued scaling of charge-based devices because of controllability limits of threshold voltage distribution. Flash density increase will continue by stacking memory layers vertically, leading to adoption of 3D Flash technology. Decrease in array efficiency due to increased interconnection and yield loss from complex processing are challenges for further reducing the cost-per-bit benefit. It is expected that more emphasis will be placed on read speed improvement of 3D Flash due to the emergence of CXL memories that are used to store AI weights.
- Ferroelectric RAM (FeRAM) is a fast, low power, and low voltage non-volatile memory (NVM) alternative and thus is suitable for radio frequency identification (RFID), smart card, ID card, and other embedded applications. Processing difficulty limits its wider adoption. If developed to maturity, this may serve as a low-power and fast, Flash-like memory.
- STT-MRAM is now mostly considered not as a standalone memory but an embedded memory. STT-MRAM would also be a potential solution for embedded Flash (NOR) replacement. This may be particularly interesting for low-power IoT applications. On the other hand, for other embedded systems applications using higher memory density, NOR Flash is expected to continue to dominate, since it is still substantially more cost-effective and well established for being able to endure the printed circuit board (PCB) soldering process (at ~250°C) without losing its preloaded code.
- 3D cross-point memory has been demonstrated for the storage class memory (SCM) to improve I/O throughput and reduce power and cost. However, in the past two years the 3D XP memory has been withdrawn from the market and is currently not available. Instead, recent attentions are steered toward a similar “selector only memory” (SOM). According to recent publications, SOM operates on a bi-directional mode, and uses only the selector but not the phase-change memory element. The selector serves both as the isolation device (like in 3D cross-point array) and as the storage device. It is not clear when a commercially available SCM based on this approach will be introduced.
- High-density resistive RAM (ReRAM) development has been limited from the lack of a good selector device, since simple diodes have limited operation ranges. Recent advances in SCM memories, however, seem to have solved this bottleneck and ReRAM could make rapid progress if other technical issues, such as erratic bits, are solved.
- PCM provides a good scaling trajectory overcoming the burdens such as high variability in RRAM and low Ron/Roff ratios in MRAM.
- More convergence of embedded memories with computing is expected giving performance and energy losses by moving data from/to the memory to/from the compute, which is the so-called memory-wall problem. This will bring in compute-in-memory (CIM) and compute-near-memory (CNM) arrays to evolve for AI applications. CIM/CNM arrays will take advantage in DRAM for high-performance AI computing and in embedded NVM devices that could be integrated in the BEOL stack bringing a better area footprint for improvement of the TOPS/mm² metric.

The links to the tables of technology roadmaps for Logic Core Device, DRAM, Flash, and NVM are below:

<i>Table MM-1</i>	<i>More Moore—Logic Core Device Technology Roadmap</i>
<i>Table MM-2</i>	<i>More Moore—DRAM Technology Roadmap</i>
<i>Table MM-3</i>	<i>More Moore—Flash Technology Roadmap</i>
<i>Table MM-4</i>	<i>More Moore—NVM Technology Roadmap</i>

3. CHALLENGES

The goal of the semiconductor industry is to be able to continue to scale the technology to improve overall performance at reduced power and cost. The performance of the components and the final chip can be measured in many different ways: higher speed, higher density, lower power, form factor reduction, bill-of-material reduction, more functionality, etc. Traditionally, dimensional scaling was adequate to bring about these performance merits, but it is no longer the case. Processing modules, tools, material properties, etc., are presenting difficult challenges to continue scaling. We have identified these difficult challenges and summarized them in Table MM-5 and Table MM-6. These challenges are divided into near-term 2024-2029 (Table MM-5) and long-term 2029-2039 (Table MM-6).

3.1. NEAR-TERM CHALLENGES

Table MM-5

Difficult Challenges—Near-term

Near-Term Challenges: 2024-2029	Description
Power scaling	Voltage and capacitance scaling slow down and there is a lack of solutions for power reduction. Introduction of gate-all-around (GAA) devices is a remedy to reduce the supply voltage, but not in a sustained manner that allows continuous scaling. Power scaling is also limited because of slow-down of loading capacitance scaling, impacted by increasing portion of parasitic components with scaling. Therefore, low-κ materials, design-technology-co-optimization (DTCO), backside wafer processing for better power delivery and contact access schemes, and novel power landing vias, are needed.
Parasitics scaling	Maintaining control of increased parasitics in stacked devices. Stacked devices require high-aspect ratio contacts to access the bottom contact. This will increase both the contact resistance as well as the fringe capacitance between the gate and drain/source. Interface resistance will also require new silicidation schemes that conformally wrap the source/drain.
Cost reduction	Cost-effective area scaling through High-NA EUV, DTCO, and 3D stacking. Throughput and yield challenges necessitate a careful selection of ground rules that optimize the die cost as a significant part of cost is determined by the middle-of-line (MOL) and BEOL stack. Therefore, new process-enhanced design constructs that tighten the secondary design rules such as tip-to-tip and the P-N separation rule are necessary to allow a further shrink of the standard cell and bitcell area on top of ground rule scaling for low-cost die. Process integration of those design constructs might require new materials to allow better etch selectivity and self-deposition. Slow-down in SRAM density scaling brings disintegration of last-level-cache and/or buffer memories as a stacked die on logic.
Integration enablement for SRAM-cache applications	Bitcell scaling is slowing down because of the slow-down of the device pitch and gate pitch. New device schemes such as N-over-P stacked device (CFET) bring an opportunity to significantly reduce the SRAM area. This is enabled because of optimized layouts that eliminate the critical design rules impacting the area. Option of embedded NVM in high-performance logic. Being able to integrate most of emerging memories (e.g., MRAM) at the interconnect stack also brings an opportunity for high-density memories. However, the stack as well as the materials should be compatible with BEOL.
Interconnect scalability	Maintaining control of interconnect resistance and EM and TDDB limits. Interconnect resistance has now entered an exponential increase regime because of non-ideal scaling of the barrier for Cu and increased scattering at the surface and grain-boundary interfaces. Therefore, there is a need for new barrier materials and high-AR Cu alternative solutions. In addition to resistance scalability, TDDB is putting a limit on the minimum space between the adjacent lines for a given low-κ dielectric.

3.2. LONG-TERM CHALLENGES

Table MM-6

Difficult Challenges—Long-term

Long-Term Challenges: 2029-2039	Description
Power scaling	Power scaling — no solutions left to reduce the supply voltage because of lack of steep-subthreshold devices. Therefore, new power delivery schemes such as backside power to further scale down the power supply voltage by reducing the supply noise. New architectures are necessary to attain performance through parallelization and increased memory capacity and bandwidth are required to reduce the amount of less energy-efficient external memory accesses.
Thermal issue due to increased power density	Thermal challenges of 3DVLSI and 3D stacking. Gate-all-around (GAA) and CFET (complementary FET) devices have limited heat conductance due to confined architecture. Increased pin density due to aggressive standard cell height scaling and increased drive by stacked devices put a significant pressure on the power density.
Cost reduction with 3D integration	Managing cost, yield, and process complexity of 3D integration. Using vertical devices separated by the interconnect significantly increases the wafer cost and the number of masks (i.e., process complexity) adding pressure to the defectivity (e.g., D0) control. Architecture needs to be refined for reducing the interconnect complexity between tiers as well as simplified integration and function per tier (e.g., I/O in one tier, SRAM in another tier, etc.).
Integration of non-Cu metallization to replace Cu	Adoption of high-AR non-Cu interconnects for low-resistance, meeting EM/TDDB, and temperature budget compatibility with devices used in 3D integration. Introduction of backside wafer processing for power distribution will require careful assessment of material and thermal considerations as this metallization will be near logic devices.

4. TECHNOLOGY REQUIREMENTS—LOGIC TECHNOLOGIES

4.1. GROUND RULES SCALING

The More Moore roadmap focuses on effective solutions to sustain the performance and power scaling at scaled dimensions and scaled supply voltage. Ground rule scaling drives die-cost reduction. However, this scaling increases the portion of parasitics in the total loading and brings diminishing returns of scale in performance and power scaling. Therefore, it is necessary to focus on technology scaling solutions that also scale the parasitics of device and interconnect. Ground-rule scaling needs to also enable DTCO constructs that accommodate the area reduction as well as tighten the critical design rules that limit the area scaling. Due to the rising costs and process complexity of multiple patterning, EUV is used as a remedy to pattern-tight ground rules in fewer process steps. The projected roadmap of ground rules as well as device architectures is shown in Table MM-7. Evolution of ground rules is shown in Figure MM-2. There is not yet a consensus on the node naming across different foundries and integrated device manufacturers (IDMs); however, the projected rules give an indication of technology capabilities in line with the PPAC requirements. Key parameters in the ground rules are the gate pitch, metal pitch, fin pitch, gate length, and 3D tier stacking capability, which are important factors in core logic area scaling.

Table MM-7

Device Architecture and Ground Rules Roadmap for Logic Devices.

Note: GxxMxx/Tx notation refers to Gxx: contacted gate pitch, Mxx: tightest metal pitch in nm, Tx: number of tiers. This notation illustrates the technology pitch scaling capability. On top of pitch scaling there are other elements such as cell height, number of stacked devices, DTCO constructs, 3D integration, etc. that define the target area scaling (gates/mm²).

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fine-pitch 3D integration scheme	Stacking	Stacking	Stacking	Stacking	CFET	CFET	3DVL SI	3DVL SI	3DVL SI
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
LOGIC DEVICE GROUND RULES									
Mx pitch (nm)	32	24	24	20	18	16	14	14	14
M1 pitch (nm)	32	32	32	46	44	44	42	42	42
M0 pitch (nm)	24	22	22	20	18	16	14	14	14
Gate pitch (nm)	48	48	48	46	44	44	42	42	42
Lg: Gate Length - HP (nm)	16	14	12	12	12	10	10	10	10
Lg: Gate Length - HD (nm)	18	14	12	12	12	10	10	10	10
Channel overlap ratio - two-sided	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
Spacer width (nm)	6	6	5	5	4	4	4	4	4
Spacer k value	3.5	3.3	3.0	3.0	2.7	2.7	2.7	2.7	2.7
Contact CD (nm) - finFET, LGAA	20	22	26	24	24	26	24	24	24
Device architecture key ground rules									
Device lateral pitch (nm)	24	26	24	24	22	22	22	22	22
Device gate vertical overhang (nm)	15	15	15	10	10	10	10	10	10
Device height (nm)	48	57	72	64	75	62	60	53	53
FinFET Fin width (nm)	5.0								
Footprint drive efficiency - finFET	4.21								
Lateral GAA vertical pitch (nm)		18.0	17.0	16.0	15.0	14.0	13.0	12.0	12.0
Lateral GAA (nanosheet) thickness (nm)		6.0	6.0	6.0	5.0	4.0	4.0	4.0	4.0
Number of vertically stacked nanosheets on one device		3	4	4	3	3	3	3	3
LGAA width (nm) - HP		30	20	12	30	25	20	15	12
LGAA width (nm) - HD		15	10	6	15	12	10	8	8
Footprint drive efficiency - lateral GAA - HP		4.41	5.47	4.80	4.57	4.24	4.00	3.68	3.43
Device effective width (nm) - HP	101.0	216.0	208.0	144.0	210.0	174.0	144.0	114.0	96.0
Device effective width (nm) - HD	101.0	126.0	128.0	96.0	120.0	96.0	84.0	72.0	72.0
PN separation width (nm)	45	40	30	20					
CFET PN separation vertical space (nm)					30	20	20	15	15

Acronyms used in the table (in order of appearance): LGAA—lateral gate-all-around-device (GAA), CFET (Complementary Field Effect Transistor), 3DVL SI—fine-pitch 3D logic sequential integration.

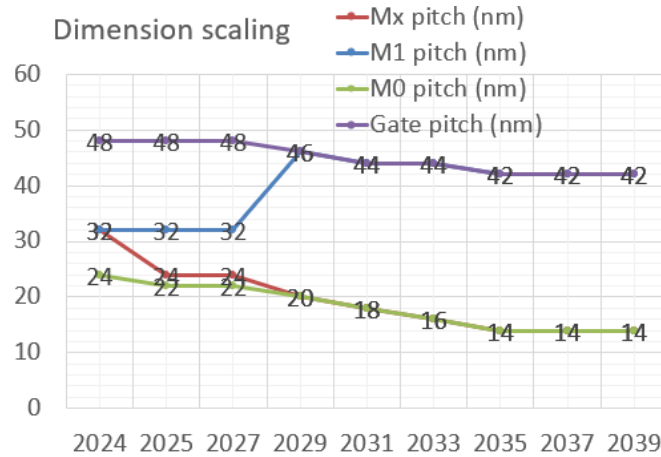


Figure MM-2

Projected scaling of key ground rules.

Ground rule scaling alone is not adequate to scale the cell height. It is necessary to bring the design scaling factor into practice [5]. For example, standard cell height will be further reduced by scaling the number/width of active devices in the standard cell as well as scaling the secondary rules such as tip-to-tip, extension, P-N separation, and minimum area rules. Similarly, the standard cell width can be reduced by focusing on critical design rules such as fin termination at the edge device, etc., and enabling structures such as contact-over-active [6][7]. Also, the contact structure needs to be carefully selected to reduce the risk of increased current density at the junctions. Also providing power from the backside metallization will free up the standard cell from the power rails and therefore further reduce the cell height. It is also expected that beyond 2031 P and N devices could be integrated on top of each other allowing a further reduction [8]. This trend in standard cell scaling is shown in Figure MM-3. The evolution of backside metallization connection schemes is

shown in Figure MM-4. In the first introduction (in 2025) backside metal is projected to be contacted to the buried rail and and/or local interconnect, while from 2027 onwards direct backside contact is projected to be contacted to the terminals (source, drain, and gate) of device to further reduce the cell height [6][9].

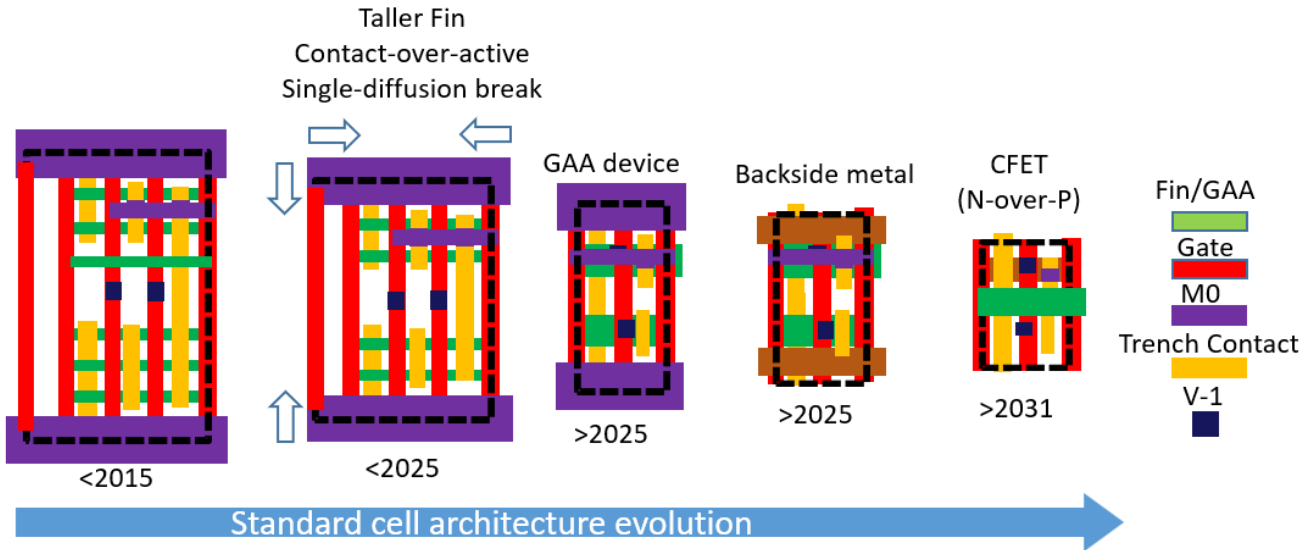


Figure MM-3 Scaling of standard cell height and width through fin depopulation and device stacking.

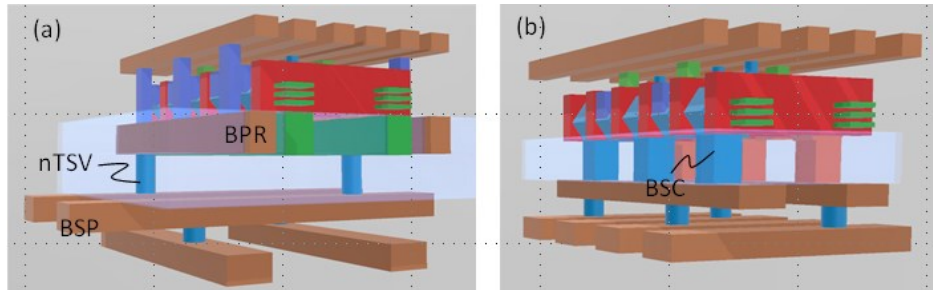


Figure MM-4 Backside metallization schemes: 1) Nano TSV (e.g., power via), 2) Direct backside contact.

It is projected that physical channel length would saturate around 12nm due to worsening electrostatics while gate pitch reduction reserving sufficient width for the device contact, providing acceptable parasitics. This drawback of pitch scaling has been compromised with dual-gate-pitch processing where relaxed pitch devices are used for high-performance cells while tight-pitch devices are used for high-density cells.

After 2035 there is no room for 2D geometry scaling, where 3D very large scale integration (VLSI) of circuits and systems using sequential/stacked integration approaches will be necessary. This is due to the fact that there is no room for contact placement as well as worsening performance as a result of gate pitch scaling and metal pitch scaling. 3D VLSI expects to bring PPAC gains for the target node as well as to pave ways for heterogeneous and/or hybrid integration. The challenge of such integration in 3D is how to partition the system to come up with better utilization of devices, interconnects, and sub-systems such as memory, analog, and I/O. That is why the functional scaling and/or significant architectural changes are required after 2035. Early generation of 3D VLSI is expected to start with CFET in 2031 where NMOS and PMOS are placed on top of each other.

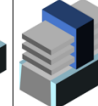
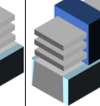
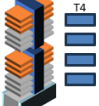
4.2. PERFORMANCE BOOSTERS

In order to maintain the scaling at low voltages, scaling in recent years focused on additional solutions to boost the performance such as the use of introducing strain to channel; stress boosters; high- κ metal gate; lowering contact resistance, and improving electrostatics [10]–[15]. This was all done in order to compensate the gate drive loss while supply voltage needs to be scaled down for high-performance mobile applications.

A roadmap overview of device architecture, key modules, and performance boosters is shown in Table MM-8.

Table MM-8

Device Roadmap and Technology Anchors for More Moore Scaling.

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic Industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fine-pitch 3D integration scheme	"3nm" Enhanced Stacking	"2nm" Stacking	"1.4nm" Stacking	"A10 eq" Stacking	"A7 eq" Stacking	"A5 eq" Stacking	"A3.5 eq" Stacking	"A2.5 eq" Stacking	"A1.5 eq" Stacking
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
LOGIC TECHNOLOGY ANCHORS									
Device technology inflection	Taller fin	LGAA	LGAA	LGAA	CFET	CFET	3D VLSI	3D VLSI	3D VLSI
Patterning technology inflection for Mx interconnect	193i, EUV DP	193i, EUV DP	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV
Specialty transistors add-ons	-	-	-	2D, IGZO	2D, IGZO	2D, IGZO	2D, IGZO	2D, IGZO	2D, IGZO
Local interconnect inflection	Self-aligned vias	Backside via	Backside contact	Backside contact	Backside contact	Backside contact	Backside contact	Backside contact	Backside contact
Process technology inflection	Channel, RMG	Lateral/Atomic Etch	Subtractive metal	Subtractive metal	Active backside	Active backside	Active backside	Active backside	Active backside
Heterogeneous stacking generation inflection	2.5D Chiplet, Mem-on-Logic	Backside Power	Backside Fabrics	Backside Fabrics	CFET	CFET	3D VLSI	3D VLSI	3D VLSI

FinFET has remained as the key device architecture that could sustain scaling until 2025. Electrostatics and fin depopulation (i.e., increasing fin height while reducing number of fins at unit footprint area) remain as the two effective solutions to improve performance. Parasitics improvement is expected to continue as a major knob for performance improvement as a result of tightening design rules. It is forecasted that the parasitics will remain as a dominant term in the performance of critical paths. For reduced supply voltage, a transition to GAA structures such as lateral nanosheets would be necessary to sustain the gate drive by improved electrostatics. Monolithic 3D integration would allow integration of devices on top of each other. Scaling focus will shift from single-thread performance gain to power reduction and then evolve onto highly-parallel 3D architectures allowing low V_{dd} operation and more functions embedded at unit cube volume. Squeeze of active area portion will make other design rules bottleneck in design scaling such as area reserved for the power rail is expected to be buried through backside rail below the device contact layer to allocate this for additional intra-cell routing.

While device architectures are seeing changes, subsequent modules are expected to also evolve. These may include: channel material evolving from Si to SiGe, Ge, 2D materials; contact module evolving from silicides to novel materials providing lower Schottky barrier height (SBH), wrap-around contact integration schemes to increase the contact surface area. Below is a list of these schemes.

4.2.1. Transition to new device architectures

As mentioned earlier finFET is likely to sustain until 2025. In 2025 a transition to lateral GAA devices is needed due to the limits of limited single fin performance and fin-width scaling (saturating the L_{gate} scaling to sustain the electrostatics control). Parasitic capacitance mitigation, effective drive width (W_{eff}), and replacement metal gate (RMG) integration continue to be the challenge to sustain GAA adoption across multiple nodes [16][17]. Projected evolution of device architectures is shown in Figure MM-5. It is projected that complementary FET (CFET) will be the subsequent evolution of lateral GAAs in the 3D form where N devices will be stacked over P devices [8].

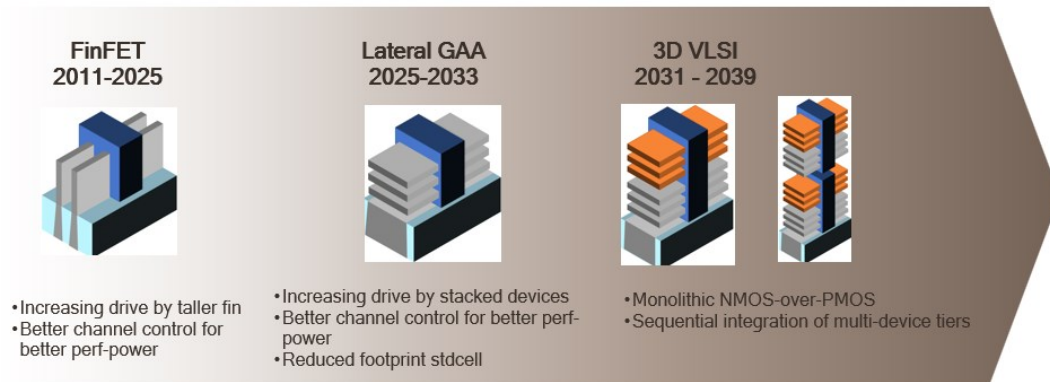


Figure MM-5

Evolution of device architectures in the IRDS More Moore roadmap.

4.2.2. Starting Substrate and high-mobility channels

Bulk silicon will still remain the mainstream substrate while silicon-on-insulator (SOI) and strain-relaxation-buffer (SRB) will be used to support better isolation (e.g., RF co-integration) and defect-free integration of high-mobility channels, respectively.

High-mobility materials such as Ge and III-V bring promise in increasing drive current by means of an order of magnitude increase in intrinsic mobility. It is likely that high-mobility channels will occur in the form of chiplets dedicated to high-performance functions, such as high-speed IOs, high-current analog drivers, RF, photonics devices, power management, etc, which does not need to follow aggressive dimensional scaling. Improved performance and enablement of new features in the total system needs to be weighed against the cost, determined by substantial investment in new tools and fab infrastructure. On the other hand, increasing the number of vertical stacked nanosheets employing high mobility channels allow very high performance at reduced footprint [18].

4.2.3. Strain engineering

Strain engineering has been used as one of the most effective solutions in the last decade, as illustrated for the 32nm node and earlier. However, the effect of those stressors may not extrapolate intuitively into newer nodes as a result of new device architectures and 3D device schemes. With the scaling down of gate pitch, SiGe on the source/drain epitaxial (S/D EPI) contact and strain relaxation buffer (SRB) remain as effective boosters to scale mobility more than double on top of high-mobility channel material. SiGe channel for PMOS and strained Si channel for NMOS has been successfully demonstrated on a 7nm CMOS platform using SRB and on gate-all-around devices. Other strain engineering techniques also contain gate stressor and ground plane stressors, which adopt the beneficiary vertical stress components for NMOS.

4.2.4. Reducing parasitic device resistance

Controlling source/drain series resistance within tolerable limits will become increasingly difficult. Due to the increase of current density, the demand for lower resistance with smaller dimensions at the same time poses a great challenge. It is estimated that in current technologies, series resistance degrades the saturation current by 40% or more. External resistance impact on the drive current is expected to become worse with the gate pitch scaling. In addition, increasing interconnect resistance by scaling is expected to necessitate much lower resistance values for the device contact and therefore more emphasis is given to increase the contact surface area by using the wrap-around contact schemes.

4.2.5. Reducing parasitic device capacitance

Parasitic capacitance between gate and source/drain terminal of the device is expected to increase with technology scaling. In fact, this component is getting more important than channel-capacitance-related loading whenever the standard cell context is considered and even more elevated in the GAA structures as a result of unused space between stacked devices. There is a need to focus on low- κ spacer materials and integration of inner spacer schemes in the GAA device is necessary to block the gate di-electric interface facing the source/drain at the unused space of GAA device. Those schemes still need to provide good reliability and etch selectivity for S/D contact formation [19]. Also, there are significant limits in increasing finFET or lateral GAA device AC performance by increasing the height of the device (fin/nanosheet stack). Energy per switch vs. delay relationship seems to quickly saturate and then decline with increasing device height, requiring to reduce the vertical space between the GAA sheets. Scaling trend of key parasitic improvements is shown Figure MM-4.

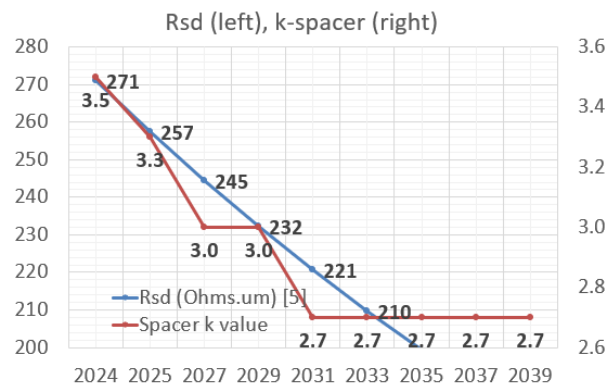


Figure MM-6 Scaling trend of device S/D access resistance (R_{sd}) and k -value of device spacer. Note [5]: R_{sd} is the total parasitic series resistance (source plus drain) per micron of MOSFET width. These values include all

components such as accumulation layer, spreading resistance, sheet resistance, and contacts. It is assumed that there is 5% improvement per node cycle (2 years or 3 years).

4.2.6. Increasing drive per footprint

Lateral GAA devices enable a higher drive at unit footprint (by enabling drive in the third dimension) if device pitch can be aggressively scaled while increasing the number of stacked GAA devices [19]. This will then increase drive at unit footprint but bringing a trade-off between fringing capacitance between gate and contact, and series resistance. For the GAA devices this is accomplished by the increasing the number of GAA sheets while reducing the width of each GAA sheet. Complementary FET (CFET) will also scale the drive per footprint by increasing the number of devices as in GAA. This will substantially increase the amount of devices at unit footprint.

4.2.7. Improving electrostatics

LGAA devices bring better electrostatics than finFET by providing a gate control from all-sides of device channel. Since the devices are stacked on top of each other, the spacing between the devices needs to be kept smaller to reduce the parasitic capacitance between the source/drain and gate while still having adequate space of the gate dielectric and V_t -tuning work function metals deposition. Junction formation engineering, EOT scaling and density of interface traps (D_{it}) reduction are also needed to maintain the electrostatics control in the channel.

4.2.8. Improving device isolation

Besides the channel leakage induced by electrostatics, there are potentially other leakage sources such as sub-fin leakage or punchthrough current. This leakage current flows through the bottom part of the fin and the bottom GAA device from source to drain. Ground plane doping and bottom dielectric isolation below the channel would potentially solve this leakage problem.

4.2.9. Reducing process and material variations

Reducing variability would further allow supply voltage (V_{dd}) scaling. Controlling channel length and channel thickness are important to maintain the electrostatics in the channel. This would require, for example, controlling the profile of the fin and lithography processes to reduce the CD uniformity (CDU), line width roughness (LWR), and line edge roughness (LER). Dopant-free channel and low-variability work-function metals would reduce the variations in the threshold voltage. Gate stack passivation might also be needed to reduce the interface-related variations as well as maintaining the electrostatics and mobility.

4.2.10. Beyond CMOS for application-specific/complimentary functions

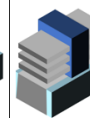
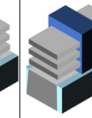
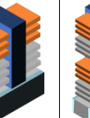
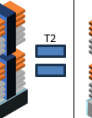
MOSFET scaling will likely become ineffective and/or very costly for the complementary SoC functions, such as memory selector, power header/footer switch, cross-bar switch, etc. Completely new, non-CMOS types of logic devices and maybe even new circuit architectures are potential solutions (see the Beyond CMOS chapter for detailed discussions). Such solutions ideally can be integrated onto the Si-based platform to take advantage of the established processing infrastructure, as well as being able to include Si devices, such as memories, onto the same chip. Even early adoption of Beyond CMOS technology and/or computing are likely to be adopted around 2029 by ferroelectric-FET, BEOL oxide transistors, IGZO, and/or 2D materials for ultra-low power applications and also memristors for neuromorphic applications [20].

4.3. PERFORMANCE-POWER-AREA (PPA) SCALING

The projected roadmap for the electrical specifications of logic core device is listed in Table MM-9. This edition of More Moore roadmap includes both logic and analog specifications of More Moore platform device. Analog specifications are derived from the device targets of a logic device but this would potentially require relaxation of contacted poly pitch on the same wafer to allow longer channel lengths. There would also be considerations such as reliability and matching where performance targets need to be derated in an effort to meet those concurrent goals, e.g. increasing overdrive voltage by stacking devices.

Table MM-9

Projected Electrical Specifications of Logic Core Device

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic Industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fine-pitch 3D integration scheme	"3nm" Enhanced Stacking	"2nm" Stacking	"1.4nm" Stacking	"A10 eq" Stacking	"A7 eq" CFET	"A5 eq" CFET	"A3.5 eq" 3DVL SI	"A2.5 eq" CFET-3D	"A1.8 eq" 3DVL SI
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
LOGIC DEVICE ELECTRICAL SPECS									
Power Supply Voltage - Vdd (V)	0.70	0.65	0.60	0.60	0.60	0.60	0.60	0.60	0.60
Subthreshold slope (mV/dec) - HP (mV/dec)	82	72	70	70	70	70	70	70	70
Subthreshold slope (mV/dec) - HD (mV/dec)	75	67	67	65	65	65	65	65	65
Capacitive equivalent thickness (CET) (nm) [2]	1.00	1.00	0.90	0.90	0.90	0.90	0.90	0.90	0.90
Vt,sat at Ioff=10nA/um - HP (mV)	156	165	164	152	164	164	158	151	146
Vt,sat (mV) at Ioff=100pA/um - HD (mV) [3][4]	288	271	276	260	266	265	261	257	257
Average effective mobility of N/P-MOS (cm ² /V.s)	125	100	80	80	60	60	60	60	60
Rsd (Ohms.um) [5]	271	257	245	232	221	210	199	189	180
Ballisticity Injection velocity (cm/s)	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06
Vdsat (V) - HP	0.092	0.101	0.108	0.108	0.144	0.120	0.120	0.120	0.120
Vdsat (V) - HD	0.104	0.101	0.108	0.108	0.144	0.120	0.120	0.120	0.120
Average Ion (uA/Perimeter Weff-um) of N/P-MOS at Ioff=10nA/um - HP [6]	874	787	759	797	768	800	827	858	885
Average Ion (uA/sheet or fin) of N/P-MOS at Ioff=10nA/um - HP [7]	88	170	158	115	161	139	119	98	85
Average Ion (uA/Perimeter Weff-um) of N/P MOS at Ioff=100pA/um - HD [8]	644	602	546	589	565	595	614	635	646
Average Ion (uA/sheet or fin) of N/P MOS at Ioff=100pA/um - HD [9]	65	130	114	85	119	104	88	72	62
Cch, total (fF/um ²) - HP/HD [8]	34.52	34.52	38.35	38.35	38.35	38.35	38.35	38.35	38.35
Gate height over fin (nm)	20	15	10	10	10	10	10	10	10
Cch (fF/um) - HP [8]	0.44	0.39	0.37	0.37	0.37	0.31	0.31	0.31	0.31
Cch (fF/um) - HD [8]	0.50	0.39	0.37	0.37	0.37	0.31	0.31	0.31	0.31
CV/I (ps) - FO3 load, HP [9]	1.06	0.96	0.87	0.83	0.86	0.69	0.67	0.64	0.62
I/CV (1/ps) - FO3 load, HP [10]	0.94	1.04	1.15	1.20	1.16	1.45	1.50	1.55	1.60
Energy per switching [CV2] (fJ/switch) - FO3 load, HP	0.65	0.49	0.40	0.40	0.40	0.33	0.33	0.33	0.33

An important speed metric for the transistor is the intrinsic delay (CV/I) where C includes the gate capacitance plus the gate fringing capacitances. These fringing capacitances have been found to be larger than the intrinsic capacitance over the channel region. This requires a modeling of parasitic components in the device. The ratio of total fringing capacitances to the gate capacitance over the channel is increasing with scaling.

In order to capture the behavior of a wireloaded datapath to connect the device parameters to SoC, we use a ring-oscillator-based circuit model where each stage is implemented with a D4 inverter driving a wireload with its branches driving three D4 inverters for the high-performance devices.

In this datapath model the delay of each stage is approximated by the Elmore expression given below [3]:

$$T_{del} = 0.69 \cdot R_{dr} \cdot C_{int} + (0.69 \cdot R_{dr} + 0.38 \cdot R_w) \cdot C_w + 0.69 \cdot (R_{dr} + R_w) \cdot C_{out}$$

where R_{dr} is the resistance of driver, C_{int} is the capacitance seen at the output of driver, R_w is the wire resistance, C_w is the wire capacitance, and C_{out} is the load capacitance due to the gates connected to the load. For logic technologies beyond 10nm the dominant term is typically found to be $R_w \cdot C_{out}$ [3]. This means that increasing the driver strength does not help if there is no improvement in the parasitic resistance of interconnect and/or a reduction in the parasitic loading of standard cell.

Projected standard cell and bitcell layout characteristics (e.g., number of active devices, W_{eff} , etc) are shown in Table MM-10.

Performance scaling across six nodes spanning from 2024 to 2039 is projected to have a mild increase (10-15% on a node-to-node basis) for datapaths with wireload resistance slowing down the performance scaling. We also take into account the wirelength reduction as function of area scaling translating into the reduction of wire-related loading capacitance and resistance. Wirelength is expected to further reduce as a result of 3DVL SI after 2031.

Energy per switching reduction is forecasted to become limited. This is mostly achieved by fin/device depopulation, which also enables the cell height reduction bringing a scaling of wire and cell related capacitances. We also consider that DT CO constructs such as contact-over-active, single diffusion break, tighter spacing between N and P, etc., as described in [6], will further reduce the standard cell height and width. After 2031 3DVL SI monolithic/sequential integration (full-scale 3DVL SI) would further maintain the scaling of the number of functions per unit cube.

Table MM-10 Logic cell and bitcell architecture.

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic Industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fine-pitch 3D integration scheme	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Logic device structure options	Stacking	Stacking	Stacking	Stacking	CFET	CFET	CFET 3D	CFET 3D	CFET 3D
Backside structure options	finFET	Backside via	Direct contact	Decap-ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
LOGIC CELL AND BITCELL ARCHITECTURE									
SRAM bitcell area (μm^2)	0.018	0.015	0.014	0.013	0.008	0.008	0.007	0.007	0.007
SRAM bitcell area portion	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60
HD SRAM area density (Mbits/mm ²)	33	40	43	45	77	77	81	81	81
NAND2 active devices per PUI/PD for HP	2	1	1	1	1	1	1	1	1
NAND2 active devices per PUI/PD for HD	1	1	1	1	1	1	1	1	1
Power delivery structure	Frontside via	Backside via	Backside via	Backside via	Backside via	Backside via	Backside via	Backside via	Backside via
Power rail width (nm)	24	22	22	20	18	16	14	14	14
Cell height (nm) - HP	192	144	144	120	108	96	84	84	84
Cell height (nm) - HD	160	120	96	80	63	56	49	49	49
Cell height in 1 μm routing tracks - HD	5.00	5.00	4.00	4.00	3.50	3.50	3.50	3.50	3.50
Routing utilization target	0.80	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
NAND2-eq gate density (Mgates/mm ²) - HP	16	23	23	29	33	37	45	45	45
NAND2-eq gate density (Mgates/mm ²) - HD	19	27	34	43	57	64	76	76	76
Effective drive width of D4 ND2 cell (nm) - HP	808	864	832	576	840	696	576	456	384
Effective drive width of D1 ND2 cell (nm) - HD	101	126	128	96	120	96	84	72	72

4.4. SYSTEM-ON-CHIP (SOC) PPA METRICS

Due to standard cell and bitcell density improving on a node-to-node basis, it is possible to integrate more functions in a given SoC footprint. The footprint for mobile SoC integration is assumed to increase across generations because of newly added functions exceeding the scaling reduction. Therefore, the amount of memory as well as graphical processing unit (GPU) processors and neural processing units (NPU) follow the density scaling of SRAM and standard cell, respectively, and if the trend for more parallel architectures continues. Thanks to advances in DTCO, lateral nanosheets, followed by device-over-device stacking (e.g. P-over-N) and 3D VLSI, SoC footprint scaling factor for the same function can still be maintained. In this roadmap edition we also added more details on machine learning blocks such as the MAC (multiply-accumulate) block clock frequency and number of MACs in SoC.

Integration capacity of logic technology is shown in Figure MM-6 (amount of NAND2-equivalent standard cell density as well as bitcell density).

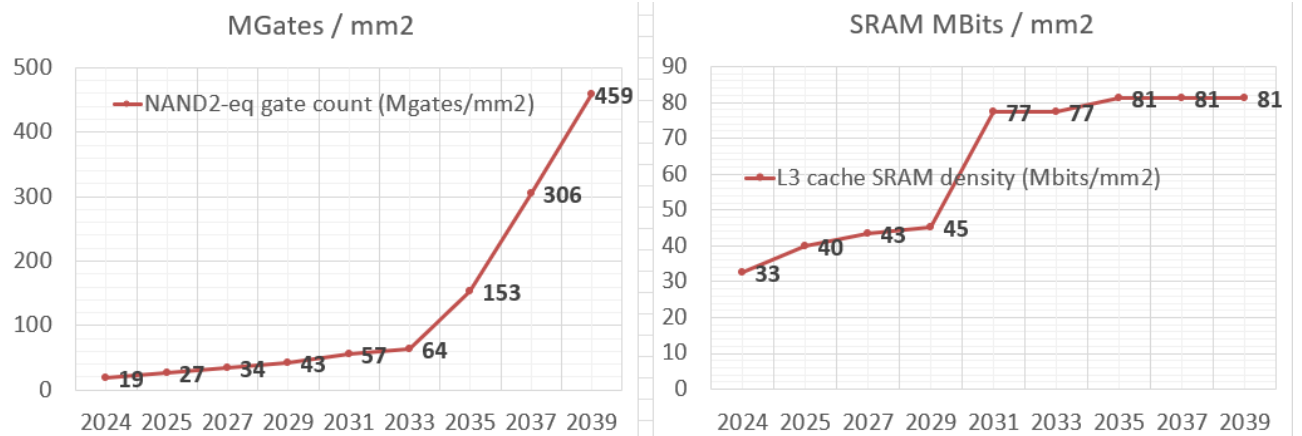
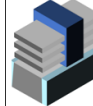
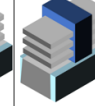
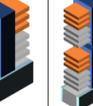


Figure MM-7 NAND2-equivalent standard cell count (left) and 1T1-bitcell (right) scaling.

Projected power and performance scaling of SoC is given in Table MM-11.

Table MM-11 Area, Power, and Performance Scaling of SoC

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fin-pitch 3D integration scheme	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Logic device structure options	finFET	Stacking	Stacking	Stacking	CFET	CFET	3DVL SI	3DVL SI	3DVL SI
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
LOGIC TECHNOLOGY KEY CHARACTERISTICS									
Vdd (V)	0.70	0.65	0.60	0.60	0.60	0.60	0.60	0.60	0.60
Gate length (nm)	16	14	12	12	12	10	10	10	10
Number of stacked tiers [1]	1	1	1	1	1	1	2	4	6
Number of stacked nanosheets in PMOS/NMOS [1]	-	3	4	4	3	3	3	3	3
Cell height (nm) - HD	160	120	96	80	63	56	49	49	49
NAND2-eq gate count (Migates/mm ²)	19	27	34	43	57	64	153	306	459
L3 cache SRAM density (Mbits/mm ²)	33	40	43	45	77	77	81	81	81
Analog + IO scaling	1.00	0.90	0.81	0.73	0.66	0.59	0.53	0.48	0.43
SoC footprint scaling	1.00	0.77	0.67	0.59	0.42	0.39	0.28	0.24	0.23
#GPU cores in SoC - at integration capacity	36	51	64	80	106	119	285	571	856
#CPU cores in SoC - at throughput target, #CPUx/Max	12	19	31	62	74	134	215	383	680
#MAC units in SoC - at capacity	16384	23211	29013	36330	48230	54259	129926	259851	389777
CPU frequency (GHz)	3.10	3.39	3.51	2.94	4.17	3.93	4.18	3.99	3.82
CPU frequency at constant power density (GHz)	3.10	3.14	4.14	3.02	4.07	3.60	2.00	1.13	0.79
MAC frequency (GHz)	1.80	1.97	2.04	1.71	2.42	2.28	2.42	2.32	2.22
MAC power scaling	1.00	0.76	0.60	0.55	0.50	0.47	0.38	0.32	0.29
MAC Power density scaling	1.00	1.08	0.85	0.97	1.03	1.09	2.08	3.55	4.81
MAC TOPS	1.0	1.6	2.0	2.1	4.0	4.2	10.7	20.4	29.3
MAC TOPS/mm ² scaling	1.0	2.2	3.5	4.7	11.7	13.9	84.7	323.8	696.6
MAC TOPS/W scaling	1.0	1.3	1.7	1.8	2.0	2.1	2.7	3.1	3.5
MAC TOPS/mm ² * TOPS/W	1.0	2.9	5.9	8.5	23.4	29.5	225.6	1014.0	2411.7

Clock frequency is projected to mildly improve because of increasing parasitics and limited gate drive ($V_{gs}-V_t$) as function of scaling. After 2035 3D-VLSI will help to reduce the parasitics through the wirelength reduction and through the split of cells in 3D. Also, thermal (increasing power density) constraints reduce the average frequency if the chip needs to be operated at constant power density. If nothing is done for the mitigation of thermal issues, the CPU needs to be throttled more frequently to maintain the same power density. The rate of power reduction tends to flatten because of slow-down in supply voltage (V_{dd}) and slow-down of capacitance scaling towards the end of roadmap. Impact of those trends in frequency, area, and energy to the system metrics such as area-efficient performance (TOPS/mm²) and energy-efficient performance (TOPS/W) is shown in Figure MM-8.

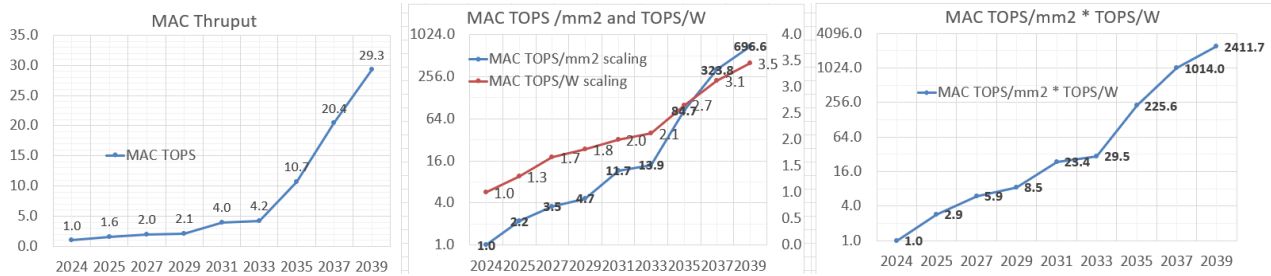


Figure MM-8 Scaling projection of MAC throughput, MAC TOPS/mm², and TOPS/W, and system area-and-power constrained performance metric (TOPS/mm²*TOPS/W).

4.5. INTERCONNECT TECHNOLOGY REQUIREMENTS

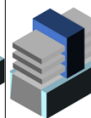
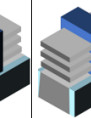
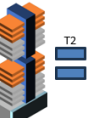
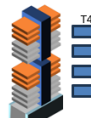
The most difficult challenge for interconnects is the introduction of new materials that meet the wire conductivity requirements, reduce dielectric permittivity, and meet reliability requirements. As for the conductivity, the impact of size effects on interconnect structures must be mitigated. Future effective κ requirements preclude the use of a trench etch stop for dual damascene structures. Dimensional control is a key challenge for present and future interconnect technology generations and the resulting difficult challenge for etch is to form precise trench and via structures in low- κ dielectric material to reduce variability in resistance-capacitance (RC). The damascene scheme used for integration requires tight control of pattern, etch, and planarization. To extract maximum performance, interconnect structures cannot tolerate variability in profiles without producing undesirable RC degradation. These dimensional control requirements place new demands on high-throughput imaging metrology for measurement of high aspect ratio structures. New metrology techniques are also needed for inline monitoring of adhesion and defects. Larger wafers and the need to limit test wafers will drive the

adoption of more in-situ process control techniques. Table MM-12 highlights and differentiates the top key challenges while Table MM-13 shows the interconnect scaling roadmap.

Table MM-12 Interconnect Difficult Challenges

Critical Challenges	Summary of Issues
Materials—Mitigate impact of size effects in interconnect structures	Line and via sidewall roughness, intersection of porous low- κ voids with sidewall, barrier roughness, and copper surface roughness will all adversely affect electron scattering in copper lines and cause increases in resistivity.
Metrology—Three-dimensional control of interconnect features (with its associated metrology) will be required	Line edge roughness, trench depth and profile, via shape, etch bias, thinning due to cleaning, CMP effects. The multiplicity of levels, combined with new materials, reduced feature size and pattern dependent processes, use of alternative memories, optical and RF interconnect, continues to challenge.
Process—Patterning, cleaning, and filling at nano-dimensions	As features shrink, etching, cleaning, and filling high aspect ratio structures will be challenging, especially for low- κ dual damascene metal structures and DRAM at nano-dimensions.
Complexity in Integration—Integration of new processes and structures, including interconnects for emerging devices	Combinations of materials and processes used to fabricate new structures create integration complexity. The increased number of levels exacerbates thermomechanical effects. Novel/active devices may be incorporated into the interconnect.
Practical Approach for 3D—Identify solutions that address 3D interconnect structures and other packaging issues	Three-dimensional chip stacking circumvents the deficiencies of traditional interconnect scaling by providing enhanced functional diversity. Engineering manufacturable solutions that meet cost targets for this technology is a key interconnect challenge.
Growing gap between area-efficient signaling and power distribution in the local interconnects	Increasing trade-off between area-efficient wiring within the cell versus low-resistance power delivery to the standard cell, necessitating decoupling of power rails from the signal routing such as introduction of buried power rails.

Table MM-13 Interconnect Roadmap for Scaling

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fine-pitch 3D integration scheme	~3nm" Enhanced	"2nm"	"1.4nm"	"1.1nm"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Logic device structure options	finFET	Stacking	Stacking	Stacking	CFET	CFET	3D VLSI	3D VLSI	3D VLSI
Backside structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Platform device for logic	finFET	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
									
FRONTSIDE INTERCONNECT TECHNOLOGY									
Number of Mx layers	2	2	2	2	2	2	2	2	2
Number of P40 layers	2	2	2	2	2	2	2	2	2
Number of P80 and <P720 layers	8	9	10	11	12	12	12	12	12
Number of P720 layers	2	2	2	2	2	2	2	2	2
Number of wiring layers - M1+Mx+ >P40	15	16	17	18	19	19	19	19	19
Mx - tight-pitch interconnect resistance (Ohms/um)	300	475	475	475	475	475	475	475	475
Mx - tight-pitch interconnect capacitance (aF/um)	270	270	270	270	270	270	270	270	270
Vx - tight-pitch interconnect via resistance (Ohms/via)	50.0	53.0	64.0	64.0	64.0	64.0	64.0	64.0	64.0
MP80 - 80nm pitch interconnect resistance (Ohms/um)	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3
MP80 - 80nm pitch interconnect capacitance (aF/um)	198	198	198	198	198	198	198	198	198
VP80 - 80nm pitch interconnect via resistance (Ohms/via)	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Aspect ratio - M0, M1, Mx, MP80, MP720	1.5-2.5	1.5-2.5	1.5-4.0	1.5-6.0	1.5-6.0	1.5-6.0	1.5-6.0	1.5-6.0	1.5-6.0
Power rail layer	M0	M0	Backside	Backside	Backside	Backside	Backside	Backside	Backside
Power rail material	Co, W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru
Metallization - M0	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru
Barrier - M0	0.5nm TaNCo	0.5nm TaNCo	Cu	Barrierless	Barrierless	Barrierless	Barrierless	Barrierless	Barrierless
Metallization - M1, Mx	Cu	Cu	Cu	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo
Barrier metal - M1, Mx	1.5nm TaNCo	1.0nm TaNCo	0.5nm TaNCo	SICOH	SICOH (2.20-2.55)	SICOH (2.20-2.55)	SICOH (2.20-2.55)	SICOH (2.20-2.55)	SICOH (2.20-2.55)
Di-electrics k value - M0, M1, Mx	SICOH (2.70-3.20)	SICOH (2.70-3.20)	SICOH (2.70-3.20)	SICOH (2.70-3.20)	Airgap (1.0)	Airgap (1.0)	Airgap (1.0)	Airgap (1.0)	Airgap (1.0)
Metallization - >MP40	Cu	Cu	Cu	Cu	Cu	Cu	Cu	Cu	Cu
Di-electrics k value - >MP40	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)

4.5.1. Conductor

Copper (Cu) is expected to remain to be the preferred solution for the interconnect metal, at least until 2029 while non-Cu solutions (e.g. Co and Ru) are expected to be used for the local interconnect (M0). On the other hand, due to limits of electromigration, the local interconnect (middle-of-line (MOL)), M1, and Mx levels will embed non-Cu solutions such as Cobalt (Co), particularly for the via, due to its better integration window to fill the narrow trenches on top of the EM performance as well as its lower resistance compared to Cu at scaled dimensions. Although a resistivity increase due to electron scattering in Cu or higher bulk resistivity in non-Cu solutions (e.g., Co) are already apparent, a hierarchical wiring approach such as scaling of line length along with that of the width still can overcome the problem.

4.5.2. Barrier Metal

Cu wiring barrier materials must prevent Cu diffusion into the adjacent dielectric but also must form a suitable, high quality interface with Cu to limit vacancy diffusion and achieve acceptable electromigration lifetimes. Ta(N) is a well-known industry solution. Although the scaling of Ta(N) deposited by plasma vapor deposition (PVD) is limited, other nitrides such as Mn(N) that can be deposited by chemical vapor deposition (CVD) or atomic layer deposition (ALD) have recently attracted attention. As for the emerging materials, self-assembled monolayers (SAMs) are researched as the candidates for future generation.

4.5.3. Inter-metal Dielectrics (IMD)

Reduction of the IMD κ value is slowing down because of problems with manufacturability. The poor mechanical strength and adhesion properties of low- κ materials are obstructing their incorporation. Delamination and damage during CMP are major problems at early stages of development, but for mass production, the hardness and adhesion properties needed to sustain the stress imposed during assembly and packaging must also be achieved. Difficulties associated with the integration of highly porous ultra-low- κ ($\kappa \leq 2$) materials become clearer, and air-gap technologies is the alternative path to lower the inter-layer capacitance. As the emerging materials, metal organic framework (MOF) and carbon organic framework (COF) could be advocated.

4.5.4. Reliability—Electromigration

An effective scaling model has been established in the earlier editions of roadmap where it assumes that the void is located at the cathode end of the interconnect wire containing a single via with a drift velocity dominated by interfacial diffusion. The model predicts that lifetime scales with w^2h/j , where w is the linewidth (or the via diameter), h the interconnect thickness, and j the current density. Whereas the geometrical model predicts that the lifetime decreases by half for each new generation, it can also be affected by small process variations of the interconnect dimensions. J_{max} (maximum equivalent DC current density) and JEM (DC current density at the electromigration limit) are limited by the interconnect geometry scaling. J_{max} increases with scaling due to reduction in the interconnect cross-section and increase in the maximum operating frequency. The practical solutions to overcome the lifetime decrease in the narrow linewidths have been discussed actively over the past years. Recent studies show an increasingly important role of grain structure in contributing to the drift velocity and thus the EM reliability beyond the 45nm node. Process solutions with Cu alloys seed layer (e.g., Al or Mn) have shown to be an optimum approach to increase the lifetime. Other approaches are the insertion of a thin metal layer (e.g., CoWP or CVD Co) between the Cu trench and the dielectric SiCN barrier and the usage of the short length effect. The short length effect has effectively been used to extend the current carrying capability of conductor lines and has dominated the current density design rule for interconnects.

4.5.5. Reliability—Time Dependent Dielectric Breakdown

Basically, the dielectric reliability can be categorized according to the failure paths and mechanisms as shown in Figure MM-9. While a large number of factors and mechanisms have already been identified, the physical understanding is far from complete. For instance, it is necessary to correctly account for LER, voltage dependence, etc in modeling TDDDB lifetime that directly impacts the estimate of V_{max} (or minimum dielectric spacing).

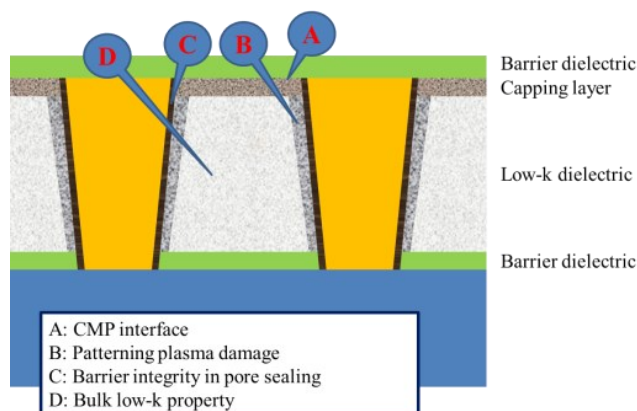


Figure MM-9 Degradation paths in low- κ damascene structure.

4.6. 3D INTEGRATION

Every logic generation needs to add new functions in each node to keep unit price constant (to preserve profit margins). This is getting more difficult due to the following challenges:

- Fewer functions left on board/system to co-integrate
- Heterogeneous cores specialized per function - specialized performance improvement requirements needed per each dedicated core
- Off-package memories - costly to co-integrate with logic, technology not compatible with baseline CMOS (where wafer/die-level stacking might be needed), limited bandwidth and energy efficiency to transport data from memory to logic

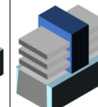
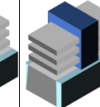
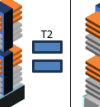
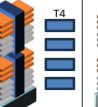
Die cost reduction has been enabled so far by concurrent scaling of gate pitch, metal pitch, and cell height scaling. This is expected to continue until 2031 and this slow-down has been accompanied by fine-pitch 3D stacking assembly such as ubump stacking and hybrid bonding [21]. Cell height scaling will continue to be pursued by DTCO constructs in cell and physical design. However, this scaling route is expected to be more challenged by diminishing electrical/system benefits and also by diminishing area-reduction/\$ at SoC level. Therefore, it is necessary to pursue 3D integration routes such as device-over-device stacking, fine-pitch layer transfer, and/or monolithic 3D (or sequential integration). These pursuits will maintain system performance and power gains while potentially maintaining the cost advantages such as treating expensive non-scaled components somewhere else and using the best technology fit per tier functionality. 3D stacking routes should factor in know-good-die sorting and test methodologies to improve the stacking yield where wafer-to-wafer stacking would require a very yielding process on each of the stacked wafers because of testing and wafer sorting challenges. Going more heterogeneity in die stacking would require careful planning how tiers are partitioned, for instance having a smaller I/O die on top logic die would require a lot of 2D routing in the logic die to fanin the connections from the corresponding logic block on the logic tier to the I/Os in the IO tier above. This routing would itself introduce the some area penalty in the logic tier. The overall trade-off should also include the assembly/stacking yield and additional wafer process steps such as TSV, wafer thinning, Cu pad/uBump processing.

3DVLSI can be routed either at gate or transistor levels. 3DVLSI offers the possibility to stack tiers enabling high-density contacts at the tier level (up to several million vias per mm^2). The partitioning at the gate level allows IC performance gain due to wire length reduction while partitioning at the transistor level by stacking nFET over pFET (or the opposite), enabling the independent optimization of both types of transistors (customized implementation of channel material/substrate orientation/channel and raised source/drain strain, etc. [16]) while enabling reduced process complexity compared to a planar co-integration, for instance the stacking of III-V nFETs above SiGe pFETs [27]. These high mobility transistors are well suited for 3DVLSI because their process temperatures are intrinsically low. There is a significant momentum on integrating device-on-device stacking (e.g. N device over P) to decouple the channel engineering (e.g. Ge channel for PMOS) for better performance. Better performance of higher tiers achieved by a freedom to select a better substrate should however factor in the potential degradation of performance due to processing them at lower temperature budget compared to the devices at the most bottom tier.

In order to address the transition from 2D to 3DVLSI, the following generations are projected in the roadmap:

- Die-to-wafer and wafer-to-wafer stacking as shown in Table MM-14
 - Approach: Fine-pitch dielectric/hybrid bonding and/or flip-chip assembly
 - Opportunities: Reducing bill-of-materials on the system, heterogeneous integration, high-bandwidth, and low latency memory on logic
 - Challenges: Design/architecture partitioning, power distribution network, thermal
- Device-on-device (e.g., P-over-N stacking)
 - Approach: Monolithic/Sequential integration
 - Opportunities: Reducing 2D footprint of standard cell and/or bit cell
 - Challenges: Minimizing interconnect overhead is key between N&P enabling low-cost
- Adding logic 3D SRAM and/or DRAM stack (embedded/stacked)
 - Approach: Sequential integration and/or wafer transfer
 - Opportunities: 2D area gain, better connection between logic and memory enabling system latency, bandwidth, and capacity gains.
 - Challenges: Solving the thermal budget of interconnect at the lower tier if stacking approach is used, Revisiting the cache hierarchy, memory IO/bus, and application requirements, power and clock distribution
- Adding Analog and I/O
 - Approach: Sequential integration and/or wafer transfer
 - Opportunities: Giving more freedom to design and allowing integration of high-mobility channels, pushing non-scaling components to another tier, IP re-use, scalability, IO voltage enablement in advanced nodes
 - Challenges: Thermal budget, reliability requirements, power and clock distribution
- 3D VLSI: Clustered functional stacks
 - Approach: Sequential integration and/or wafer transfer
 - Opportunities: Complementary functions other than CMOS replacement such as neuromorphic, high-bandwidth memory or pure logic applications incorporating new data-flow schemes favoring 3D connecting. Application examples include image recognition in neuromorphic fabric, wide-IO sensor interfacing (e.g., DNA sequencing, molecular analysis), and highly parallel logic-in-memory computations.
 - Challenges: Architecting applications where low energy at low frequency and highly parallel interfaces could be utilized, mapping applications to non-Von Neumann architectures.

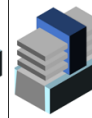
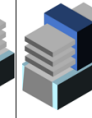
Table MM-14 3D stacking and backside metallization technology ground rules.

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Logic industry "Node Range" Labeling	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Fine-pitch 3D integration scheme	Stacking	Stacking	Stacking	Stacking	CFET	CFET	3DVLSI	3DVLSI	3DVLSI
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
STACKING, BACKSIDE INTERCONNECT and 3DVLSI									
Die-to-wafer hybrid bonding pad pitch (um)	10	10	7	5	3	3	3	3	3
Die-to-wafer uBump pitch (um)	40	25	25	15	15	15	15	15	15
Stacking wafer thickness (um)	40	40	30	30	20	20	20	20	20
3DVLSI (layer transfer or monolithic) tier-to-tier via pitch (nm)	-	-	-	80	63	56	49	49	49
Number of backside metals	-	4	4	5	5	6	6	6	6
Backside/via contact pitch (nm)	-	120	32	27	21	19	16	16	16
First backside metal pitch (nm)	-	192	32	27	21	19	16	16	16
Backside functions	-	PDN, IO	PDN, IO, Clock	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus

4.7. DEFECTIVITY REQUIREMENTS

More Moore scaling necessitates an increase in the number of metallization layers, therefore an increase in the mask count if no advancement is done in the patterning technology. The expected transition from the 193i lithography to EUV will potentially save masks. However, the mask count is expected to escalate after 2031 because of increased need for the metallization and repeated masks used for the front-end-of-line (FEOL) and middle-of-line (MOL) integration for 3D integration. This will in turn increase the process complexity, therefore the defectivity (D0) requirements. The required D0 level is expected to significantly scale down (Table MM-15). Known-good stack and/or repairable/fault-tolerant logic topologies are needed beyond 2031 because of increasing process complexity.

Table MM-15 Defectivity Requirements.

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fine-pitch 3D integration scheme	~3nm Enhanced	2nm	1.4nm	1.1nm	A7 eq	A5 eq	A3.5 eq	A2.5 eq	A1.8 eq
Logic device structure options	finFET	Stacking	Stacking	Stacking	CFET	CFET	3DVL SI	3DVL SI	3DVL SI
Backside structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Platform device for logic	finFET	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
DEFECTIVITY TARGETS									
SoC footprint area target - AI Inference (mm ²)	90	108	130	156	187	187	187	187	187
Critical area portion in a single tier	75%	75%	75%	75%	75%	75%	75%	75%	75%
Maskcount target including stacked tiers	75	81	84	87	89	90	145	255	365
Defectivity Bose-Einstein D0 target (defects/inch ²)	0.056	0.043	0.034	0.028	0.023	0.028	0.023	0.020	0.014
Defectivity Poisson D0 target (defects/cm ²)	0.29	0.27	0.26	0.25	0.24	0.30	0.25	0.21	0.15
Process complexity exponent	28.0	30.2	31.4	32.5	33.2	33.6	54.1	95.2	136.3
Repairable logic portion	0%	0%	0%	0%	0%	20%	40%	60%	60%
Wafer sort yield (%) - sequential 3D assumed for T>1	85%	85%	85%	85%	85%	85%	85%	85%	85%

4.8. DEVICE RELIABILITY

Reliability is an important requirement for all users of integrated circuits. The challenge of realizing the required levels of reliability is increasing due to (1) scaling, (2) new materials and devices, (3) more demanding mission profiles (higher temperatures, extreme lifetimes, high currents), and (4) increasing constraints of time and money. These reliability challenges will be exacerbated by the need to introduce multiple major technological changes in a brief period. Interactions between changes can increase the difficulty of understanding and controlling failure modes. Furthermore, having to deal simultaneously with several significant issues will tax limited reliability resources.

Reliability requirements are highly application dependent. For most customers, current overall chip reliability levels (including packaging reliability) need to be maintained over the next fifteen years despite the reliability risk inherent in massive technological changes. However, there are also niche markets that require reliability levels to improve. Applications that require higher reliability levels, harsher environments, and/or longer lifetimes are more difficult than the mainstream compute and mobile applications.

4.8.1. Device reliability difficult challenges

Table MM-14 indicates the top near-term reliability challenges. The first near-term reliability challenge concerns failure mechanisms associated with the MOS transistor. The failure could be caused by breakdown of the gate dielectric or degradation of device parameters, like threshold voltage and leakage current, change beyond the acceptable limits. The time to failure is decreasing with scaling. Depending on the circuit it may take more than one soft breakdown to produce an IC failure, or the circuit may function for longer time until the initial degradation spot has progressed to a "hard" failure. Threshold voltage-related failure is primarily associated with the negative bias temperature instability observed in p channel transistors in the inversion state and the analogous positive bias temperature instability in n channel transistors. Burn-in options to enhance reliability of end-products may be impacted, as it may accelerate negative bias temperature instability (NBTI) shifts.

ICs are used in a variety of different applications. There are some special applications for which reliability is especially challenging. First, there are applications in which the environment subjects the ICs to stresses much more than found in typical consumer or office applications. For example, automotive, military, and aerospace applications subject ICs to extremes in temperature and shock. In addition, aviation and space-based applications also have a more severe radiation environment. Furthermore, applications like base stations require IC's to be continuously on for tens of years at elevated temperatures, which makes accelerated testing of limited use. Second, there are important applications (e.g., implantable electronics, safety systems) for which the consequences of an IC failure are much greater than in mainstream IC

applications. In general, scaled-down ICs are less “robust” and this makes it harder to meet the reliability requirements of these special applications. Memories, energy-harvesting, and energy-storage devices exhibit their specific degradation modes, which may be different from those of transistors, in particular, abrupt breakdowns with no signs of preceding degradation. New computing paradigms such as neuromorphic and quantum computing impose additional requirements on stability/drift of device characteristics.

At the heart of reliability engineering is the fact that there is a distribution of lifetimes for each failure mechanism. With low failure rate requirements, we are interested in the early range of the failure time distributions. There has been an increase in process variability with scaling (e.g., distribution of dopant atoms, chemical mechanical polishing (CMP) variations, and line-edge roughness). At the same time the size of a critical defect decreases with scaling. These trends will translate into an increased time spread of the failure distributions and, thus, a decreasing time to first failure. We need to develop reliability engineering software tools (e.g., screens, qualification, and reliability-aware design) that can manage the increase in variability of the device physical properties, and to implement rigorous statistical data analysis to quantify the uncertainties in reliability projections. The use of Weibull and log-normal statistics for analysis of breakdown reliability data is well established, however, the shrinking reliability margins require more careful attention to statistical confidence bounds to quantify risk. This is complicated by the fact that new failure physical mechanisms, for instance, correlated defect generation, may lead to significant deviations from Weibull distribution, making error analysis non-straightforward. Statistical analysis of several reliability processes such as bias temperature instability (BTI) and hot carrier degradation is not currently standardized in practice but may be needed for accurate modeling of circuit failure rate.

The single long-term reliability difficult challenge concerns novel, disruptive changes in devices, structures, materials, and applications. For such disruptive solutions there is at this moment little, if any, reliability knowledge (as least as far as their application in ICs is concerned). This will require significant efforts to investigate, model (both a statistical model of lifetime distributions and a physical model of how lifetime depends on stress, geometries, and materials), and apply the acquired knowledge (new building-in reliability, designing-in reliability, screens, and tests). It also seems likely that there will be less-than-historic amounts of time and money to develop these new reliability capabilities. Disruptive material or devices therefore lead to disruption in reliability capabilities, and it will take considerable resources to develop those capabilities.

Table MM-16 Device Reliability Difficult Challenges

Near-Term 2024-2029	Summary of issues
Reliability due to material scaling, process, and structural changes, and novel applications.	• TDDDB, negative BTI (NBTI), positive BTI (PBTI), hot carrier injection (HCI), random telegraphic noise (RTN) in scaled non-planar devices. • Gate to contact breakdown. • Increasing statistical variation of intrinsic failure mechanisms in scaled non-planar devices. • 3D device structure reliability challenges • Reduced reliability margins drive need for improved understanding of reliability at circuit operation conditions. • Reliability of embedded electronics in extreme or critical environments (medical, automotive, grid...)
Long-Term 2029-2039	Summary of issues
Reliability of novel devices, structures, and material stacks.	• Understand and control the failure mechanisms associated with a new combined set of materials and device structures. • Shift to system level reliability perspective with unreliable devices. • Muon induced soft error rate

4.8.2. Device reliability potential solutions

The most effective way to meet requirements is to have complete built-in-reliability and design-for-reliability solutions available at the start of the development of each new technology generation. This would enable finding the optimum reliability/performance/power choice and would enable designing a manufacturing process that can consistently have adequate reliability. Unfortunately, there are serious gaps in these capabilities today and these gaps are likely to grow even

larger in the future. The penalty will be an increasing risk of reliability problems and a reduced ability to push performance, cost, and time-to-market.

It is commonly thought that the ultimate nanoscale device will have a high degree of variation and high percentage of non-functional devices right from the start. This is viewed as an intrinsic nature of devices at the nanoscale. As a result, it will not be possible any longer for designers to consider a ‘worst case’ design window, because this would jeopardize the performance of the circuits too much. To deal with it, a complete paradigm change in circuit and system design will therefore be needed. While we are not there yet, the increase in variability is clearly already a reliability problem that is taxing the ability of most manufacturers. This is because variability degrades the accuracy of lifetime projection, forcing a dramatic increase in the number of devices tested. The coupling between variability and reliability is squeezing out the benefit of scaling. At some point, before the end of the roadmap, the cost of ensuring each one of the transistors in a large integrated circuit to function within specification may become too high to be practical. As a result, the fundamental philosophy of how to achieve product reliability may need to be changed. This concept is known as resilience, the ability to cope with stress and catastrophe. One potential solution would be to integrate so-called solutions and monitors in the circuits that are sensing circuit parts that are running out of performance and then during runtime can change the biasing of the circuits. Such solutions need to be further explored and developed. Circuits that can dynamically reconfigure themselves to avoid failing and failed devices (or to change/improve functionality) will be needed.

The growing complexity of a reliability assessment due to proliferation of new materials; gate stack compositions tuned to a variety of specific applications; as well as shorter cycle for process development, may be alleviated to some degree by greater use of the physics-based microscopic reliability models, which are linked to material structure simulations and consider degradation processes on atomic level. Such models, a need for which is slowly getting wider recognition, will reduce our reliance on statistical approaches, which is both expensive and time consuming, as discussed above. These models can provide additional advantages because they can be incorporated in compact modeling tools with relative ease and require only a limited calibration prior to being applied to a specific product.

Some slight changes may already be underway quietly. A first step may be simply to fine-tune the reliability requirements to trim out the excess margin, even having product-specific reliability specifications. More sophisticated approaches involve fault-tolerant design, fault-tolerant architecture, and fault-tolerant systems. Research in this direction has increased substantially. However, the gap between device reliability and system reliability is large. There is a strong need for device reliability investigation to address the impact on circuits. The recent increase in using circuits such as SRAM and ring oscillators to look at many of the known device reliability issues is a good sign, as it addresses both the issues of circuit sensitivity as well as variability. More device reliability research is needed to address the circuit and system aspects. For example, most of the device reliability studies are based on quasi-DC measurements. There is no substantial research on the impact of degradation on devices at circuit operation speed. This gap in measurement speed makes modeling the impact of device degradation on circuit performance difficult and risky.

In the meantime, we must meet the conventional reliability requirements. That means an in-depth understanding of the physics of each failure mechanism and the development of powerful and practical reliability engineering tools. Historically, it has taken many years (typically a decade) before the start of production for a new technology generation to develop the needed capabilities (R&D is conducted on characterizing failure modes, deriving validated, predictive models and developing design for reliability and reliability TCAD tools.) The ability to qualify technologies has improved, but there still are significant gaps.

For the reliability capabilities to catch up requires a substantial increase in reliability research-development-application and cleverness in acquiring the needed capabilities in much less than the historic time scales. Work is needed on rapid characterization techniques, validated models, and design tools for each failure mechanism. The impact of new materials like alternate channel material needs particular attention. Breakthroughs may be needed to develop design for reliability tools that can provide a high-fidelity simulation of a large fraction of an IC in a reasonable time. As mentioned above, increased reliability resources will also be needed to manage the introduction of many major technological changes in a brief period.

The needs are clearly many, but a specific one is the optimal reliability evaluation methodology, which would deliver relevant long-term degradation assessment while avoiding excessive accelerated testing that may produce misleading results. This need is driven by the decreasing process margin and increasing variability, which degrades the accuracy of lifetime projection from a standard sample size. The ability to stress a large number of devices simultaneously is highly desirable, particularly for long term reliability characterization. Doing it at manageable cost is a challenge that is difficult to meet and becoming more so as we migrate to more advanced technology nodes. A breakthrough in testing technology is badly needed to address this problem.

5. TECHNOLOGY REQUIREMENTS—MEMORY TECHNOLOGIES

CMOS logic and memory together form the predominant majority of semiconductor device production. The types of memory considered in this chapter are DRAM and non-volatile memory (NVM). The emphasis is on commodity, stand-alone chips, since those chips tend to drive the memory technology. However, embedded memory chips are expected to follow the same trends as the commodity memory chips, usually with some time lag. For both DRAM and NVM, detailed technology requirements and potential solutions are considered.

5.1. DRAM

For DRAM, the main goal is to continue scaling the footprint of the 1T-1C cell, to the practical limit of 4F2. The challenges are vertical transistor structures, high- κ dielectrics to improve the capacitance density, while keeping the leakage low. In general, technical requirements for DRAMs become more difficult with scaling. In the past several years, DRAM was introduced with many new technologies (e.g., 193 nm argon fluoride (ArF) immersion high-NA lithography with double patterning technology, improved cell FET technology including fin type transistor [22], buried word line/cell FET technology [23] and so on).

Since the DRAM storage capacitor gets physically smaller with scaling, the equivalent oxide thickness (EOT) must scale down sharply to maintain adequate storage capacitance. To scale the EOT, dielectric materials having high relative dielectric constant (κ) will be needed. Therefore, metal-insulator-metal (MIM) capacitors have been adopted using high- κ ($\text{ZrO}_2/\text{Al}_2\text{O}_3/\text{ZrO}_2$) [24] as the capacitor of DRAMs having the ground rules between 48nm and 30nm half-pitch. And this material evolution and improvement are continued until 20 nm HP and ultra high- κ (perovskite $\kappa > 50 \sim 100$) material are released. Also, the physical thickness of the high- κ insulator should be scaled down to fit the minimum feature size. Due to that, capacitor 3-D structure will be changed from cylinder to pillar shape.

On the other hand, with the scaling of peripheral CMOS devices, a low-temperature process flow is required for process steps after formation of these devices. This is a challenge for DRAM cell processes that are typically constructed after the CMOS devices are formed, and therefore are limited to low-temperature processing. The DRAM peripheral device requirement can relax Ioff but demands more Ion of low standby power (LSTP) devices. But, in the future, high- κ metal gate will be needed for sustaining the performance [25].

The other important topic is the migration from 6F2 to 4F2 cell. As the half-pitch scaling becomes very difficult, it is impossible to sustain the cost trend. The most promising way to keep the cost trend and increasing the total bit output by generation is changing the cell size factor (a) scaling (where $a = [\text{DRAM cell size}]/[\text{DRAM half pitch}]^2$). Currently 6F2 ($a = 6$) is the most common. For example, vertical cell transistor is required but still a couple of challenges are remaining. Other alternative is the use of 3D DRAM.

All in all, maintaining sufficient storage capacitance and adequate cell transistor performance are required to keep the retention time characteristic in the future. And their difficult requirements are increasing to continue the scaling of DRAM devices and to obtain the bigger product size (i.e., >16 Gb). In addition to that, if efficiency of cost scaling becomes poor in comparison with introducing the new technology, DRAM scaling will be stopped, and 3D cell stacking structure will be adopted, or a new DRAM concept will be adopted. 3D cell stacking and new concept DRAM are discussed but there is no clear path for further scaling beyond the 2D DRAM.

5.2. NVM—FLASH

There are several intersecting memory technologies that share one common trait—non-volatility. The requirements and challenges differ according to the applications, ranging from RFIDs that only require Kb of storage to high-density storage of hundreds of Gb in a chip. Nonvolatile memory may be divided into two large categories—Flash memories (NAND Flash and NOR Flash), and non-charge-based-storage memories. Nonvolatile memories are essentially ubiquitous, and a lot of applications use embedded memories that typically do not require leading edge technology nodes. The More Moore nonvolatile memory tables only track memory challenges and potential solutions for leading edge standalone parts.

Flash memories are based on simple one transistor (1T) cells, where a transistor serves both as the access (or cell selection) device and the storage node. At this time Flash memory serves more than 99% of applications.

When the number of stored electrons reaches statistical limits, even if devices can be further scaled and smaller cells achieved, the threshold voltage distribution of all devices in the memory array becomes uncontrollable and logic states

unpredictable. Thus memory density cannot be increased indefinitely by continued scaling of charge-based devices. However, effective density increase may continue by stacking memory layers vertically.

The economy of stacking by completing one device layer then another and so forth is questionable. As depicted in Figure MM-10 [26], the cost per bit starts to rise after stacking several layers of devices. Furthermore, the decrease in array efficiency due to increased interconnection and yield loss from complex processing may further reduce the cost-per-bit benefit of this type of 3D stacking. In 2007, a ‘punch and plug’ approach was proposed to fabricate the bit line string vertically to simplify the processing steps dramatically [26]. This approach makes 3D stacked devices in a few steps and not through repetitive processing, thus promised a new low-cost scaling path to NAND flash. Figure MM-10 illustrates one such approach. Originally coined bit-cost-scalable, or BiCS, this architecture turns the NAND string by 90 degrees from a horizontal position to vertical. The word line (WL) remains in the horizontal planes. As depicted in Figure MM-10, this type of 3D approach is much more economical than the stacking of complete devices, and the cost benefit does not saturate up to quite high number of layers.

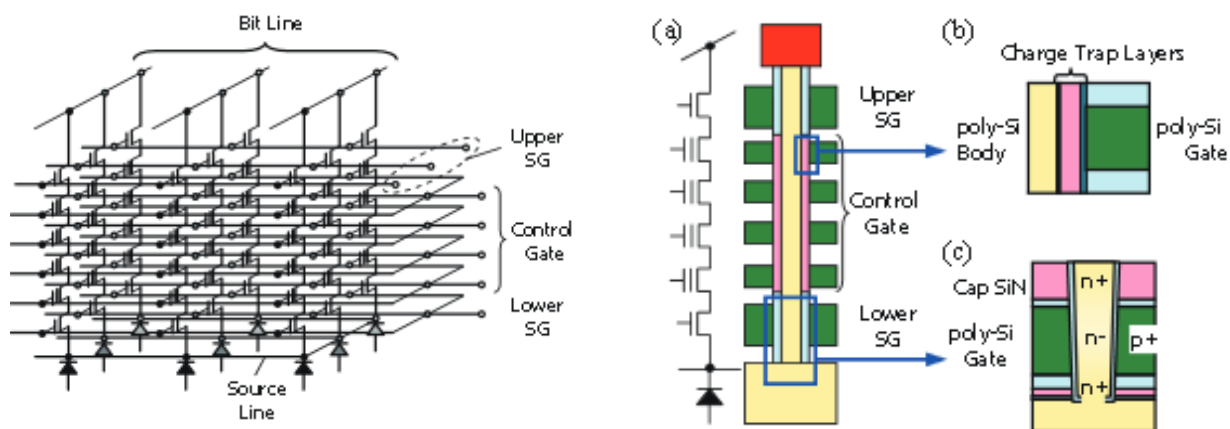


Figure MM-10 (left) A 3D NAND array based on a vertical channel architecture. (right) BiCS (bit cost scalable) – a 3D NAND structure using a punch and plug process [26].

A number of architectures based on the BiCS concept have been proposed since 2007 and several, including some that use floating gate instead of charge trapping layer for storage, have gone into volume production in the last 2–3 years. In general, all 3D NAND approaches have adopted a strategy of using much larger areal footprints than the conventional 2D NAND. The x- and y- dimensions (equivalent to cell size in 2D) of 3D NAND are in the range of 100nm and higher compared to ~15nm for the smallest 2D NAND. The much larger “cell size” is made up by stacking a large number of memory layers to achieve competitive packing density.

The economics of 3D NAND is further confounded by its complex and unique manufacturing needs. Although the larger cell size seems to relax the requirement for fine line lithography, to achieve high data rate it is desirable to use large page size and this in turn translates to fine pitched bit lines and metal lines. Therefore, even though the cell size is large metal lines still require ~20nm half-pitch that is only achievable by 193i lithography with double patterning. Etching of deep holes is difficult and slow, and the etching throughput is generally very low. Depositing many layers of dielectric and/or polysilicon, as well as metrology for multilayer films and deep holes all challenge unfamiliar territories. These all translate to large investment in new equipment and floor space and new challenges for wafer flow and yield.

The ultimate unknown is how many layers can be stacked. There seems no hard physics limit on the stacking of layers. Beyond certain aspect ratio (100:1 perhaps?) the etch-stop phenomenon, when ions in the reactive ion etching process are bent by electrostatic charge on the sidewall and cannot travel further down, may limit how many layers can be etched in one operation. However, this may be bypassed by stacking fewer layers, etching, and stacking more layers (at higher cost). Stacking many layers may produce high stress that bends the wafer and although this needs to be carefully engineered it does not seem to be an unsolvable physics limit. Even at 200 layers (at ~50nm for each layer) the total stack height is about 10μm, which is still in the same range as 10–15 metal layers for logic IC’s. This kind of layer thickness does not significantly affect bare die thickness (thinnest is about 40μm so far) yet. However, at 1000 layers the total layer thickness may cause thick dies that do not conform to the form factor for stacking multiple dies (e.g., 16 or 32) in a thin package. At this time, >250 layers are in volume production and there is optimism that 500+ layers are achievable and even 1000 layers are possible. In addition to processing challenges stacking more layers also increases the area overhead required to making

contacts to more word lines. This area overhead, plus the added processing complexity, will eventually diminish the cost benefit by adding more layers.

Renewed shrinking of the areal x-y footprint may eventually start when stacking more layers proves to be too difficult. However, such a trend is not guaranteed. If the hole aspect ratio is the limitation, shrinking the footprint would not reduce the ratio and would thus not be helpful. Furthermore, the larger cell size seems to at least partially contribute to the better performance of 3D NAND (speed and cycling reliability) compared to tight-pitch 2D NAND. Whether x-y scaling can still deliver such performance is not clear. Therefore, the roadmap projections for future generations stay the same as the current node in 2024. On the other hand, increasing the number of storage bits per memory cell, although technically challenging, seems to make progress. This is partly to take advantage that 3D NAND devices are intrinsically larger thus with more stored electrons and easier to make into more logic levels. At this time 4-bits/cell devices (QLC) are in volume production, and there is optimism that 5-bits/cell and more may become viable in the near future. Higher number of storage bits in a cell requires some trade off in performance since it takes longer to program and read, and reliability suffers when squeezing logic levels close together. Yet for many read-intensive applications such trade-offs are acceptable for lower cost.

5.3. NVM—EMERGING

Since 2D NAND Flash scaling is limited by statistical fluctuation due to too few stored charges, several non-conventional non-volatile memories that are not based on charge storage, so-called resistive or capacitive memories (ferroelectric or FeRAM, magnetic or MRAM, phase-change or PCRAM, and resistive or ReRAM), are being developed and form the category often called “emerging” memories. Even though 2D NAND is being replaced by 3D NAND (that is no longer subject to the drawback of too few electrons) some characteristics of non-charge based emerging memories, such as low voltage operation, or random access, are attractive for various applications and thus continue to be developed. These emerging memories usually have a two-terminal structure (e.g., resistor or capacitor) thus are difficult to also serve as the cell-selection device. The memory cell generally combines a separate access device in the form of 1T-1C, 1T-1R, or 1D-1R.

5.3.1. FeRAM

FeRAM devices achieve non-volatility by switching and sensing the polarization state of a ferroelectric capacitor. To read the memory state the hysteresis loop of the ferroelectric capacitor must be traced and the stored datum is destroyed and must be written back after reading (destructive read, like DRAM). Because of this ‘destructive read’ it is a challenge to find ferroelectric and electrode materials that provide both adequate change in polarization and the necessary stability over extended operating cycles. Many ferroelectric materials are foreign to the normal complement of CMOS fabrication materials, and can be degraded by conventional CMOS processing conditions. FeRAM is fast, low power, and low voltage and thus is suitable for RFID, smart card, ID card, and other embedded applications. Processing difficulty limits its wider adoption. Recently, HfO₂ based ferroelectric FET, for which the ferroelectricity serves to change the V_t of the FET and thus can form a 1T cell similar to Flash memory, has been proposed. If developed to maturity this new memory may serve as a low power and very fast Flash-like memory.

5.3.2. MRAM

Magnetic RAM (MRAM) devices employ a magnetic tunnel junction (MTJ) as the memory element. An MTJ cell consists of two ferromagnetic materials separated by a thin insulating layer that acts as a tunnel barrier. When the magnetic moment of one layer is switched to align with the other layer (or to oppose the direction of the other layer) the effective resistance to current flow through the MTJ changes. The magnitude of the tunneling current can be read to indicate whether a ONE or a ZERO is stored. Field switching MRAM probably is the closest to an ideal “universal memory” since it is non-volatile and fast and can be cycled indefinitely. Thus, it may be used as NVM as well as SRAM and DRAM. However, producing a magnetic field in an IC circuit is both difficult and inefficient. Nevertheless, field switching MTJ MRAM has successfully been made into products. The required magnetic field for switching, however, increases when the storage element scales while electromigration limits the current density that can be used to produce higher H field. Therefore, it is expected that field switch MTJ MRAM is unlikely to scale beyond 65nm node. Recent advances in “spin-transfer torque (STT)” approach where a spin-polarized current transfers its angular momentum to the free magnetic layer and thus reverses its polarity without resorting to an external magnetic field offer a new potential solution. During the spin transfer process, substantial current passes through the MTJ tunnel layer and this stress may reduce the writing endurance. Upon further scaling the stability of the storage element is subject to thermal noise, thus perpendicular magnetization materials are projected to be needed at 32nm and below. Perpendicular magnetization has been recently demonstrated.

With rapid progress of NAND Flash and the recent introduction of 3D NAND that promises to continue the equivalent scaling, the hope of STT-MRAM to replace NAND seems remote. However, its SRAM-like performance and much smaller footprint than the conventional 6T-SRAM have gained much interest in that application, especially in mobile devices that

do not require high cycling endurance, as in computation. Therefore, STT-MRAM is now mostly considered not as a standalone memory but an embedded memory [27], and is not tracked in the standalone NVM table. STT-MRAM would be a potential solution not only for embedded SRAM replacement but also for embedded Flash (NOR) replacement. This may be particularly interesting for IoT applications since low power is the most important. On the other hand, for other embedded systems applications using higher memory density, NOR Flash is expected to continue to dominate since it is still substantially more cost effective. Furthermore, Flash memory is well established for being able to endure the PCB board soldering process (at $\sim 250^{\circ}\text{C}$) without losing its preloaded code, which many emerging memories have not been able to demonstrate yet.

5.3.3. PCRAM and Crosspoint Memory

PCRAM devices use the resistivity difference between the amorphous and the crystalline states of chalcogenide glass (the most commonly used compound is $\text{Ge}_2\text{Sb}_2\text{Te}_5$, or GST) to store the logic levels. The device consists of a top electrode, the chalcogenide phase change layer, and a bottom electrode. The leakage path is cut off by an access transistor (or diode) in series with the phase change element. The phase change write operation consists of: (1) RESET, for which the chalcogenide glass is momentarily melted by a short electric pulse and then quickly quenched into amorphous solid with high resistivity, and (2) SET, in which a lower amplitude but longer pulse (usually $>100\text{ns}$) anneals the amorphous phase into low resistance crystalline state. The 1T-1R (or 1D-1R) cell is larger or smaller than NOR Flash, depending on whether MOSFET or BJT (or diode) is used. The device may be programmed to any final state without erasing the previous state, thus providing substantially faster programming throughput. The simple resistor structure and the low voltage operation also make PCRAM attractive for embedded NVM applications. The major challenges for PCRAM are the high current (fraction of mA) required to reset the phase change element, and the relatively long set time and high temperature tolerance to retain the preloaded code during solder reflow (at $\sim 250^{\circ}\text{C}$). Thermal disturb is a potential challenge for the scalability of PCRAM. However, thermal disturb effect is non-cumulative (unlike Flash memory in which the program and read disturbs that cause charge injection are cumulative) and the higher temperature RESET pulse is short (10ns). Interaction of phase change material with electrodes may pose long-term reliability issues and limit the cycling endurance and is a major challenge for DRAM-like applications. Like DRAM, PCRAM is a true random access, bit alterable memory.

The scalability of PCRAM device to $< 5\text{nm}$ has been demonstrated using carbon nanotubes as electrodes [28], and the reset current followed the extrapolation line from larger devices. In at least one case, cycling endurance of $1\text{E}11$ was demonstrated [29]. Phase change memory has been used in feature phones to replace NOR Flash since 2011, and has been in volume production at $\sim 45\text{nm}$ node since 2012, but no new product has been introduced since then. PCM memories have been also targeted in the last years as potential candidates for eFlash replacement for embedded applications [30]. In these works alloying of phase change materials of different classes allowed to obtain memory compliant to soldering reflow; however, such high temperature stability has come at the expense of slower write speed.

A 3D cross point memory (3D XP) has been reported [31]. Details are still lacking but it is speculated that the threshold switching (OTS) property of chalcogenide-based phase change material constitutes the core of the selector device responsible for the cross point cell, which was first reported in 2009 [32]. This is the first commercial realization of the widely published storage class memory (SCM) [33]. Computer systems badly need improved I/O throughput and reduce power and cost, and it is a promising candidate to change the entire memory hierarchy not only for high-end computation but for mobile systems as well. In addition, since the memory including the selector device is completely fabricated in the BEOL process it is relatively inexpensive to stack multiple layers to reduce bit cost.

3D cross point memory (3D XP) consists of a selector element made of ovonic threshold switching (OTS) (or an equivalent device) in series with a storage element. The selector device has a high ON/OFF ratio and is at OFF state at all times except when briefly turned on during writing or reading. The storage element is programmed to various logic states. Since the selector is always off, with high resistance the memory array has no leakage issue even if all storage elements are at low resistance state. During write or read operation the selector is temporarily turned on (by applying a voltage higher than its threshold voltage) and the OTS characteristic suddenly reduces its resistance to a very low value, allowing reading (or programming) current to be dominated by the resistance of the storage element. The storage element may be a phase-change material and in that case the memory cell is a phase-change RAM (PCRAM) switched by OTS. The storage element may also be a resistive memory material. Although bipolar operation makes the circuitry and operation more complicated, the array structure is very similar to that using PCRAM.

3D XP memory has entered commercial application for several years now, providing SCM function for servers and at some time also for PCs. Up to this date, however, it is not widely adopted since existing products use a proprietary interface. Yet, its unique advantage of random access and low cost making it a good choice for augmenting DRAM function in a memory

system thus its potential for wider adoption still exists. Its eventual success will depend on whether it can continue to hold a cost advantage over DRAM.

However, in the past two years the 3D XP memory has been withdrawn from the market and is currently not available. The exact reason is not announced. The commercial failure may be partially caused by the lack of common standard interface, and partly due to advances in DRAM scaling which narrows the cost differential.

Instead, recent attentions are steered toward a similar “selector only memory” (SOM). According to recent publications, SOM operates on a bi-directional mode, and uses only the selector but not the phase-change memory element. The selector serves both as the isolation device (like in 3D XP array) and as the storage device. The resistance of the selector changes by the direction of the applied voltage. SOM is still in research mode currently. It is not clear when a commercially available SCM based on this approach will be introduced.

Note that SLC 3D NAND can also serve SCM function with the proper interface, at much lower cost. But its considerably slower latency makes it more suitable for storage SCM, and 3D XP is more suitable for memory-type SCM (i.e., more DRAM-like).

5.3.4. *Resistive Memory (ReRAM)*

A large category of two-terminal devices, in which memory state is determined by resistivity of a metal-insulator-metal (MIM) structure, are being studied for memory applications. Many of these resistive memories are still in research stage and are discussed in more detail in the Beyond CMOS roadmap chapter. Because of their promise to scale below 10nm, and operate at extremely high frequencies ($< \text{ns}$) with low power consumption, the focused R&D efforts in many industrial labs in the last decade make this technology widely considered a potential successor to NAND (including 3D NAND). Being a two-terminal device, high-density ReRAM development has been limited by the lack of a good selector device. Recent advances in 3D XP memory, however, seem to have solved this bottleneck and ReRAM could make rapid progress if other technical issues such as erratic bits are solved. In addition to 3D XP array (like PCRAM-based 3D XP memory) high-density ReRAM products may be fabricated using a 2D array and small word-line (WL), and small bit-line (BL) half pitch. Furthermore, if eventually the OTS type of selector device is adopted it seems feasible to fabricate BiCS type 3D ReRAM using a transistor in the bottom and OTS selector for each ReRAM device in the 3D array, as depicted in Figure MM-11 [34]. Although no high-density ReRAM product has been introduced yet since the bottleneck of bipolar selector devices seems solved by the introduction of 3D XP memory, progress in ReRAM may be reasonably expected. Recently, however, the passion for developing high-density ReRAM seems to have dissipated. This may be due to two reasons. (1) the success of 3D NAND Flash has increased the entrance barrier and (2) difficulty in meeting the reliability requirements for large arrays. (Note that several announcements were made for successful development of ReRAM for smaller, Mb-size, arrays for embedded applications.)

In the last several years these above issues seem to doom large scale application of high-density ReRAM. The original argument that ReRAM, because it consists of thousands of atoms, is free from statistical fluctuation seems questionable now. It seems the filament that operates the ReRAM is made up by only a few atoms (ions). There seems evidence that even relatively large ReRAM device is subject to statistical fluctuation. Therefore, we stop tracking ReRAM for high-density application in this roadmap release.

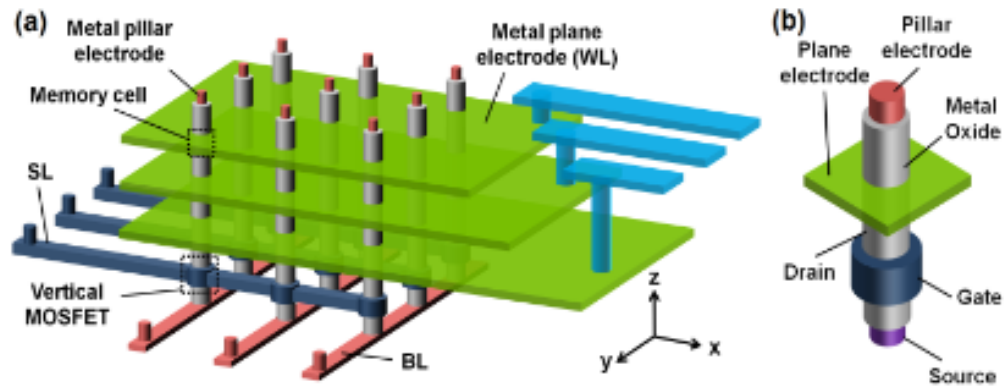


Figure MM-11 Schematic view of (a) 3D cross-point architecture using a vertical RRAM cell and (b) a vertical MOSFET transistor as the bit-line selector to enable the random access capability of individual cells in the array [34].

6. POTENTIAL SOLUTIONS

Below are the potential solutions to address the scaling challenges that were addressed in section 3 towards the targets described in section 1.2. Near-term (2024-2029) potential solutions are listed in Table MM-17 while long-term (2029-2039) potential solutions are listed in Table MM-18 .

Table MM-17 Potential Solutions—Near-term

Near-Term Potential Solutions: 2024-2029	Description
Performance	- Increasing fin height to match performance - Reducing interface contact resistance through new materials and wrap-around contact - Introducing low-κ device spacer - Reducing interconnect resistance through barrier and liner scaling
Power	- Introducing GAA architecture - Reducing device parasitics
Area and Cost	- Adoption of EUV for single and double patterning - DTCO enhancement - Introducing high-density emerging memory as cache applications

Table MM-18 Potential Solutions—Long-term

Long-Term Potential Solutions: 2029-2039	Description
Performance	- Reducing wirelength and drive-per-footprint through 3D stacking - Employing vertically stacked memory-logic sub-systems within highly parallel new computational schemes and heterogenous stacking
Power	- Increasing parallelism by the introduction of large bandwidth access to the memory - Fine-grain power gating
Area and Cost	- High regularity 3D integration/stacking with each tier adopted minimal cross-tier interconnect and integration overhead

These potential solutions are mostly targeting improvement of the PPAC value of logic technologies. It should be noted that application drivers such as 6G and extreme reality (e.g., AR/VR) bring new potential solutions for the analog/RF/sensor co-integration enablement with the use of those technology platforms. Examples include co-integration of III-V technologies with Si logic through layer transfer and/or selective growth for the enablement of products in small form factors. Si technologies, developed on low-loss SOI substrates, are expected to push the envelope of mm-wave communications where high transition frequency (F_t) and low insertion loss will be traded with a lower output power in comparison to non-Si counterparts.

Si photonics is gaining momentum in short-to-medium distance connectivity applications such as chip-to-chip communications in data server racks and back-haul network of radio access cells; and merging of energy-efficient computing with the optics in augmented reality products. Those solutions require a highly integrated interposer incorporating optical modulators, laser source, photo diodes, photonic waveguides, wave-division-multiplexors, and assembly interfaces coupling fiber to the waveguide. The requirements, challenges, and potential solutions are described in the Outside System Connectivity roadmap report.

Another growing solution is the trend of miniaturizing personalized healthcare with the co-integration of heterogeneous technologies. Those products are expected to co-integrate sensors, battery, high-endurance/high-speed non-volatile memory, RF connectivity components, and ultra-low-power processing augmented with machine learning capability in the same package. More Moore technologies are helping in this context to reduce the power consumption of those devices as well as bringing new memories (e.g., MRAM, FeRAM) required for these applications.

7. CROSS TEAMS

Through cross-functional team interaction with other International Focus Teams (IFTs), the More Moore team incorporated valuable inputs in our roadmap both in terms of requirements as well as technology capability limits:

- Systems and Architectures (SA) IFT—computational datapath/fabric such as number of CPU, GPU, and NPU cores per a given footprint as well as latency/bandwidth for data access.
- Application Benchmarking (AB) IFT—performance and energy scaling targets, chip-level power (active, static, sleep), thermal envelope
- Lithography IFT—Pitch limits of 193i and EUV lithography, CDU/LER capability, timeline of EUV in HVM adoption
- Yield IFT—unit-step related defect impact on material quality, infrastructural constraints such as CD and defect density on filtration and detection, and impact of defects on reliability.
- Metrology IFT—Extendibility of metrology of 3D devices such as lateral-GAA
- Outside System Connectivity (OSC) IFT—I/O and integration requirements for 5G and high-speed memory for data server
- Packaging Integration IFT—form factor and hetero-technology needs for mobile, 5G, and automotive
- Beyond CMOS (BC) IFT—3D memories such as RRAM and PCM, memristor for neuromorphic applications
- Environmental, Safety, Health and Sustainability (ESHS) IFT – process integration flows

8. CONCLUSIONS AND RECOMMENDATIONS

In this chapter we proposed a roadmap that could sustain More Moore scaling for concurrent enablement of performance, power, and area/cost. We identified the following inflection points:

- GAA is expected to become a mainstream device in 2025 with early introduction in 2022 and requires significant attention on the capacitance reduction to maintain performance scaling target.
- Slow-down in pitch scaling tackled with design technology co-optimization (DTCO) enables the SoC area reduction where this might require process-related dimension control is necessary besides lithography.
- We identified that 3DVLSI integration is needed beyond 2031. Thermal is becoming a significant challenge in 3D adoption and needs to revisit the architecture to get back the performance scaling through parallelization.
- We identified that significant reduction of defectivity level as well as careful split of technology and architecture across tiers are required to maximize the adoptability of 3D.
- Stacked SRAM has already become a mainstream alternative to fully integrated SRAM and/or eDRAM cache applications.

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